BUSINESS INTELLIGENCE (BI) SUPPLEMENTAL GUIDE

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The materials presented in this technical guideline provide specific supplemental information into OUM for the purposes of Business Intelligence (BI) implementation work. The BI Supplemental Guide contains the following components:

BI Overview

This component provides an overview of the concepts of Business Intelligence (BI).

BI Supplemental Activity Guidelines

The Supplemental Activity Guidelines should be used in conjunction with the standard OUM activity guidelines in order to accomplish Business Intelligence (BI) implementations.

BI Supplemental Task Guidelines for Custom BI Engagements

The Supplemental Task Guidelines for Custom BI engagements should be used in conjunction with the standard OUM tasks in order to accomplish custom BI implementations.

PDF View

You may also prefer to access this information through the **Business Intelligence (BI) Supplemental Guide (PDF View)** located in the Supplemental Guides section of the Supplemental Guidance page.

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PURPOSE

The purpose of this document is to provide an overview of BI use for providing BI implementation solutions, and to provide implementation teams with *supplemental guidance* for the successful planning and execution of BI implementation projects.

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AUDIENCE

This document is intended for practitioners who will be designing and implementing BI solutions based on the OUM (Oracle Unified Method) approach.

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HOW TO USE THIS DOCUMENT

This document contains supplemental guidance for planning and executing BI implementation projects and should be used in conjunction with and not instead of the OUM base method content.

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LIMITATIONS TO THIS DOCUMENT

This document does not describe OUM in detail – the reader must review the documentation and training materials referenced below to gain a basic understanding of each.

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RELATED DOCUMENTATION AND TRAINING

The following related documentation & training provide additional detailed guidance related to OUM implementations and are prerequisites to this document:

- OUM Method Pack – Desktop Installable or Hosted version
- OUM Training and Accreditation Program

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CONTRIBUTORS

Anthony Carpenter. Russ Gilkeson, Theresa Merenkov, Russell Pitts, Sue Ellen Strapp

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BUSINESS INTELLIGENCE (BI) OVERVIEW

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INTRODUCTION

Business Intelligence refers to technologies, applications, and practices for the collection, integration, analysis, and presentation of business information (Wikipedia, 2008). The Business Intelligence (BI) view focuses on those OUM activities and tasks that are typically leveraged for Business Intelligence implementation engagements. The OUM BI view further breaks down this BI-specific fous into the following two implementation engagements types:

- **BI Applications** - These engagements involve the delivery of projects utilizing commercial-off-the-shelf (COTS) business intelligence offerings such as EPM and other BI applications. These projects also include some amount of custom work to extend the COTS application(s), as necessary. This work is often referred to as custom extensions.
- **Custom BI** - These engagements involve custom technological BI solutions that range from large-scale data warehouses to daily operational business intelligence applications that are tailored to suit the specific needs of an enterprise. They do not include COTS offerings.

In general, Business Intelligence solutions are neither 100% custom developed (Custom BI) nor 100% product based (BI Applications). This means that BI implementation teams need to tailor their use of the OUM BI view to be fit-for-purpose. They must integrate the product or reused components with custom components resulting in the final solution. The BI Supplemental Activity Guidelines provide recommended versus optional usage guidance for each activity and its associated tasks.

Project planning, execution, and management activities are sourced directly from the from the OUM Manage focus area. This allows you to plan and manage projects in a consistent and repeatable way.

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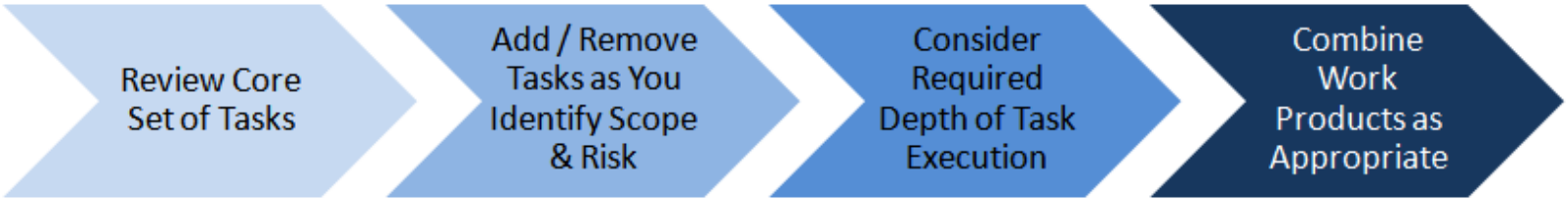
TAILORING THE VIEW FOR A SPECIFIC PROJECT

The Oracle Unified Method provides a comprehensive set of tasks and activities for conducting a broad range of software implementation projects. Each task represents a step in the process of defining and implementing a software system. OUM also recognizes that that too much method can be as risky as too little and recommends a four step approach for tailoring the process to suit a particular project. The objective of the tailoring exercise should be to identify the minimum usable subset of tasks and consolidated work products that will allow the project to be successfully completed and appropriately documented, without forcing excess "ceremony" or superfluous documentation.



1. Start with a core set of tasks. OUM and the BI view provides primary mechanisms to help project teams determine the appropriate set of tasks for a given project.

1. **OUM Implement Core Workflow** - The most basic set of tasks is defined in the OUM Implement Core Workflow. The Core Workflows till include tasks that will not be performed on every project. For instance, tasks related to Current and Future Process Models may not be required on all BI projects.
2. **Key Work Products** - OUM defines a set of key work products for each phase. This is a larger list than the Core Workflow. While these tasks are not required to be performed, they are tasks that OUM indicates may be more important to consider or may represent the culmination of work for a given phase or iteration.
3. **BI Activity Guidelines** - This view looks at OUM's tasks and activities through the prism of a BI project. The view makes recommendations regarding tasks and activities (groupings of tasks) that should be more strongly considered for inclusion. The BI Supplemental Activity Guidelines classifies activities as "Recommended" or "Optional." A recommended activity contains at least one task that has been classified as "Recommended." Optional tasks may be added when specific conditions require them. Ultimately, it is up to the project manager and team to determine the appropriate set of tasks for a specific project.

The BI view allows you to access the OUM Project Workplan. The OUM Project Workplan is a sample project workplan that can quickly reflect the BI view and then be further tailored for the specific BI engagement. It also allows the project to be managed using a plan that supports varying levels of granularity. A project manager may choose to track much of the project at the activity level, while managing a few critical activities at the task level. This allows the project manager to be more focused on project risk areas, such as work that is being performed remotely or offshore.

2. Add or remove tasks as you identify scope and risk.

Once you have determined a core set of tasks – based on the characteristics of the project consider whether it is necessary to add tasks based on scope and risk. For example, if your are obligated to provide performance testing as part of your project, you should examine the tasks and activities related to the Performance Management process. The same would be true for processes, such as Documentation, Transition, and Organizational Change Management.

3. Consider the depth to which it is necessary to execute specific tasks in a given phase or iteration.

This step may be the most important. Placing a task into an OUM iteration plan is not a binary event. Tasks are not all or nothing, but merely represent a step in the process. You must consider the depth to which you should execute tasks on your project as a whole, or within a given phase or iteration. Too much or too little depth or "ceremony" are both risky. Under the proper circumstances, spending the time to simply *consider* a task will constitute executing the task. This is an appropriate practice and is often preferable to eliminating the task altogether.

Tasks are there to remind you of the process and order of the work, not to force you to perform unnecessary work or create voluminous documentation. Also, under the proper circumstances it is may be appropriate for the output of a task - the work product - to be tacit knowledge in the head of a project team member, rather than a model or document (as long as that project team member is able to communicate that knowledge to the other members of the team, as required).

4. Combine work products into an appropriate set of consolidated work products or deliverables.

Most OUM tasks are accompanied by a work product template. Templates are provided for the convenience of the project team. OUM does not recommend that every one of these templates be used on every project, even for the tasks that appear on the project work plan. While OUM values documents and deliverables – when they are necessary – it places a higher value on working software that supports the needs of the business. It is imperative that teams make thoughtful decisions about the number and depth of the work products, especially documents, that are produced on a project.

The BI view supports the concept that work products should be pulled together into a smaller number of consolidated artifacts. This supports the *"fit for purpose"* theme upon which OUM is built. It would not be uncommon for consolidated artifacts to be identified as key deliverables for the project. These consolidated artifacts are not necessarily documents. For example, one such artifact will be the "Production-Ready System," which OUM considers to be one of the most important work products for

any project...*a working system*.

Note: Creating consolidated artifacts does not mean eliminating implementation processes. Project teams should still work through the defined process at a depth and level of ceremony that is appropriate to the project. Limiting "documentation" to a pragmatic set of consolidated artifacts helps maintain the project's focus on collaboration and the creation of working software and away from effort wasted on creating comprehensive, yet difficult-to-maintain documentation.

To best understand the philosophical approach that OUM takes to software implementation, go to Oracle's Full Lifecycle Method for Deploying Oracle-Based Business Solutions for further information on the scalability and adaptability of OUM.

By adopting OUM's iterative and incremental approach, implementation teams create a project environment that is more responsive to change – and to the deepening understanding of business requirements that occurs during a project – without sacrificing the ability to manage cost and schedule. This "think a little, do a little...think a little more, do a little more" approach means that task outputs, work products, consolidated artifacts, etc. are developed iteratively across a defined set of phases, iterations, or sprints.

OUM recommends looking at breadth before depth. For example, an early iteration the use case model might consist of a list of use cases, actors, and goals – some of which are supported fully by the COTS products and some which may not be fully supported. As the project progresses over subsequent iterations, the use cases which require custom software might be further elaborated with a brief description, a main success scenario, and, potentially, a list of exception scenarios. Ultimately, the use case model is detailed to a depth that allows the requirements for the system to be unambiguously documented. Of course, this choice of depth depends on the characteristics of the project. Co-located teams of empowered business users and implementers – who maintain a high degree of interaction and communication – require less documentation than projects with non co-located teams or projects for which the final requirements authority is held apart from the on-the-ground team.

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KEY CONCEPTS

There are several key concepts that are important to understand in order to fully appreciate the BI implementation approach for delivering an engagement using OUM.

Key Concept	Description
Business Process Models and Use Cases for Requirements Gathering	The OUM BI view employs techniques such as interviews and workshops to gain a comprehensive understanding of business requirements that are documented using business process models and use cases. Use cases complement business process models in a number of important ways. See the OUM Method Overview to understand more about the relationship between use case modeling and business process modeling.
Black-Box Use Case	Black-box use cases capture requirements at the level of observable behavior, but do not reveal the internal workings of the business or system. Most use cases are written at the black-box level to maintain the separation between requirements and design. Within this view, black-box use cases are most applicable to development of a custom BI system, but would also apply to the implementation of Custom BI and BI Applications (such as, EPM) or product-based systems that require custom extensions that support **new** data entry or reporting requirements.
White-Box Use Case	White-box use cases capture requirements that reveal the internal workings of a business or system. White-box use cases detail **how** the system will satisfy the requirements. White-box use cases are applicable on projects that will implement BI Applications, such as EPM product-based systems to document requirements regarding changes in the way the product needs to work.
Black-Box versus White-Box	It is important to understand how black-box and white-box use cases are applied to support BI projects. Black-box use cases are typically used to capture requirements related to the development of new data entry, display, and reporting mechanisms. This includes new custom data entry and report components or data entry and reporting extensions being made to a BI product (BI Applications). White box use cases are used to capture requirements related to the development or customization of internal data extraction, transformation, and loading components. These requirements typically result in the configuration or customization of existing BI Applications. When using white-box use cases, remember to keep the use cases at the level of requirements and not to delve into system design.

Domain Model	A comprehensive data model exists for BI Apps products. A domain model is used to collect data requirements for custom developed BI systems or to capture data requirements for custom extensions to BI Application products (such as EPM). The domain model is the primary input into the logical database design. OUM recommends, but does not require, the use of a UML class model for this purpose. Project teams should choose a modeling approach that is appropriate to the type of data being modeled, the skills of the designers, and the target platform. Entity-relationship diagrams (ERD's), multi-dimensional data models, or even simple data tables may also be appropriate for this purpose.

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APPLYING OUM REQUIREMENTS MODELING CONCEPTS

One key features of OUM is the utilization of use cases to capture requirements in an form that supports strong traceability of requirements throughout the lifecycle of a project. This section of the overview highlights how use cases influence the design and implementation of business intelligence solutions. By properly employing OUM's use case techniques, business analysts create requirements artifacts that are well defined, easily understood, and structured in a manner that allows them to become direct inputs to the analysis/design, implementation, and testing processes.

To effectively employ use cases on BI projects, it is essential to develop these artifacts at the correct level of granularity. Use cases that are too finely grained require are expensive to produce and provide limited marginal value to the project. When use cases that are too broadly defined, teams run the risk of missing critical requirements. OUM's use case-related task guidance and training materials provide extensive guidance related to the appropriate level of use case granularity. The depth and level of ceremony to which the use cases are developed must also be considered and appropriately adjusted to the characteristics of the project, most importantly the level of criticality of the system under discussion.

Project teams must also consider the level at which to begin use case development. OUM provides support for high-level use cases, business use cases, system use cases (black box), and sub-subsystem use cases (white-box). This choice of where to begin primarily on two factors:

- **Project Scope and Objectives**

  If the scope for the project explicitly includes examination of the business or business processes, then development of high-level use cases (BA.060) or business use cases (RA.015) may be warranted. However, if the bulk of the business analysis work was completed in preparation for the project, and the business objectives are well understood, these business-related artifacts will be inputs to the projects and do not need to be developed. Project teams should model only to the level of detail required to correctly install, configure, develop, and extend the BI system.

- **Degree of Custom Development and Customization**

  Oracle offers a comprehensive set of products to support the business intelligence and performance management needs of an enterprise. Projects to implement these products fall into three basic categories; each of which may require a different level of business analysis:

  Custom BI projects - Model the business to a level that allows the team to understand the business context in which the system will reside and provide an understanding of the purpose and scope of the system. Use the appropriate combination of business context model, business process models, and business use case model.

  Use these business artifacts as direct inputs to system use case and domain models that describe the behavior and data requirements for the *system*. Typically, only black-box use cases are employed when defining requirements for a system that has yet to be designed.

  BI Applications (COTS) projects with new data structure requirements - Model the business to a level that allows the team to understand the business context in which the system will reside and provide an understanding of the purpose and scope of the system. Limit the effort to the analysis required to define product setup parameters and identify gaps between the products' out-of-the-box functionality and the requirements of the enterprise. Use an appropriate combination of business context model, business process models, and business use case model.

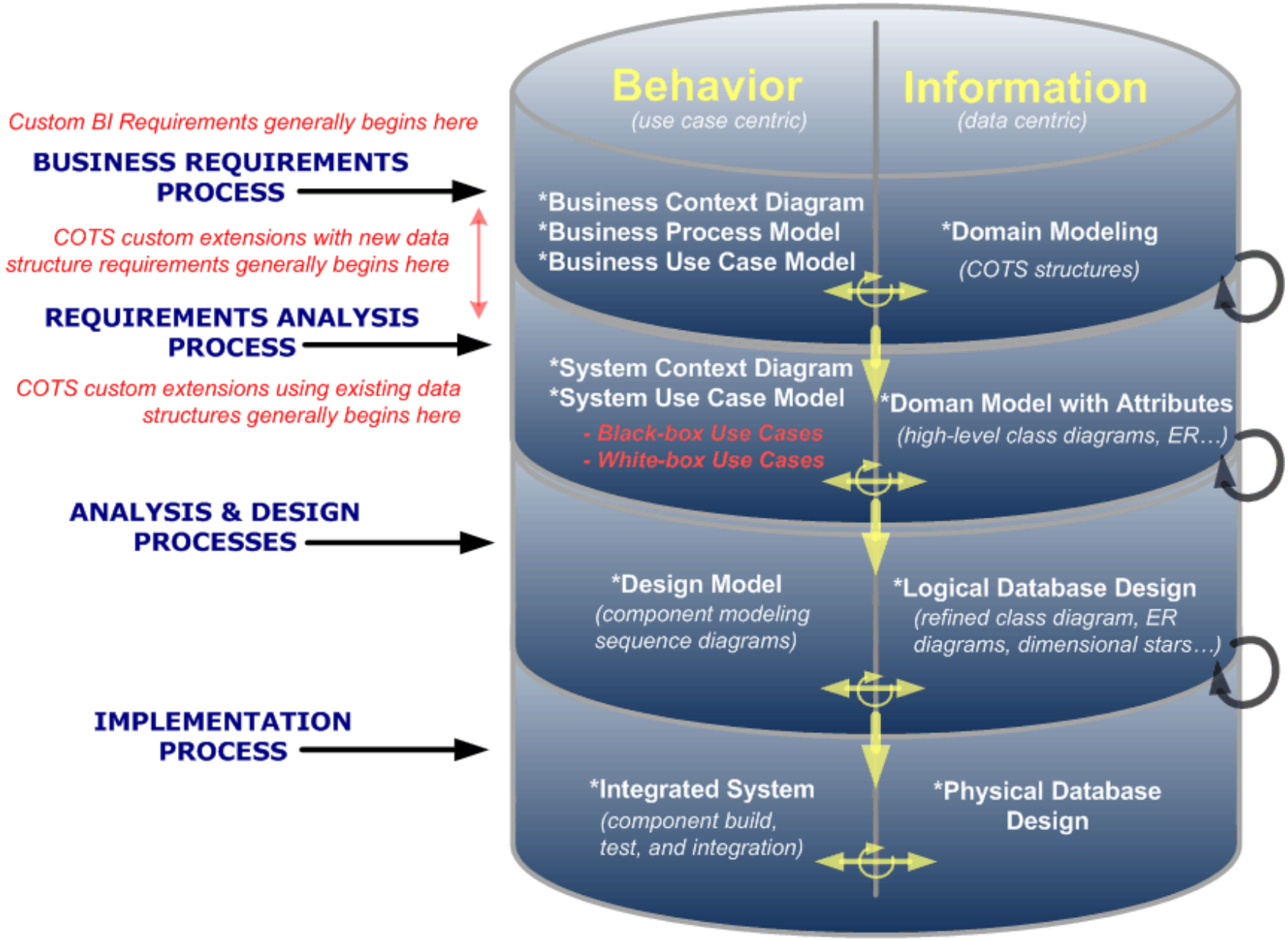
Where possible, use business process and business use case models as direct inputs to define system setup parameters. Where complex configurations are required, these artifacts are the starting point for system use cases that will, in turn, be used to define system setup parameters. Business models may also be used as direct inputs to system use case and domain models that describe the behavior and data requirements for *customizations or extensions* required to fill the gaps. Both black-box and white-box use cases may be employed, as necessary, to define the configuration and customization requirements.

BI Applications (COTS) projects using existing data structures - Model the business to a level that allows the team to understand the business context in which the system will reside and provide an understanding of the purpose and scope of the system. Limit business modeling to areas required to assist in defining product setup parameters using the appropriate combination of business context model, business process models, and business use case model.

Where possible, use business process and business use case models as direct inputs to define system setup parameters. Where complex configurations are required, these artifacts are the starting point for system use cases that will, in turn, be used to define system setup parameters. Typically, white-box use cases will be employed, as necessary, to define the configuration requirements.

In each instance, projects should limit business and system requirements modeling to the scope being addressed by the system. Teams should only develop those artifacts that are necessary to unambiguously establish the behavior and data requirements for the system.

This illustration portrays, at a high-level, the key artifacts that OUM provides for moving from business requirements into an implemented system. This includes artifacts for capturing business requirements, translating business requirements into system requirements, translating system requirements into a system design, and, ultimately, implementing that design into a working software system. The illustration shows two parallel and complementary paths that are used to model and implement the behavior and information requirements for the system.



The following table supports the illustration by providing supporting text related to activities that typically occur within each process. It should be noted that there are other OUM processes that need to be performed as well in order to deliver the solution (e.g., project management, testing, technical architecture, etc.); however the focus here is on eliciting and capturing requirements and translating those requirements into a system that will support the requirements.

It is important to note that each of these activities are interrelated and will be done incrementally over a number of iterations. This supports the *"think a little, do a little - Think a little more, do a little more "* approach that OUM embraces. This means that each of these processes is revisited and artifacts are refined as the project moves through the phases and iterations. It is also important to note that the use of prototypes may also be essential to arriving at an appropriate solution to support the specified business requirements.

		Key Artifacts		

Process	Key Purpose of Process	Typically Produced	Behavior Perspective...	Data Perspective...
Business Requirements	Define the business requirements and business needs that will be supported by the proposed system.	• Business Context Diagram • Business Process Models • Business Use Case Model	Critical to identifying new business behaviors that typically occur when new user data entry and data structures are required.	High-level domain modeling begins with reviewing existing data structures associated with the chosen COTS application products, as well as capturing entities associated with required data structures that are not supported by the chosen products.
Requirements Analysis	The business requirements are refined and translated into functional requirements for a system that will support those requirements. The business requirements are analyzed further and a System Use Case Model is developed. As the project progresses, the Use Case Model may be further elaborated, where necessary, by developing use case details that establish a more precise understanding of requirements. Use case details are especially important where significant custom development or complex customizations will be required to support those use cases. Depending upon the characteristics of the system being developed, both black-box and white-box use cases may be employed to capture the requirements, as described in the Key Concepts section above.	• System Context Diagram • System Use Case Model • System Use Case Details	Begins with defining the boundaries for the system, the actors that will interact with the system, and the information needs of those actors. **(System Context Diagram)** Moves to understanding, at a high level, the functions supported by the system and the actors that will initiate or participate in those functions. **(System Use Case Diagram)** Moves to understanding how human and system actors will use the system without understanding system's internal workings/constraints. **(System Black-Box Use Cases)** May also involve developing a detailed description of the internals of the system to support custom extensions. **(System White-Box Use Cases)**	The Domain Model is further developed and attributed based on analysis of the requirements. The results are a high-level class diagram or other data modeling artifacts such as Entity Relationship (ER) Diagram, as required to support the data modeling needs of the system.
Analysis and Design	Analysis - The system requirements are mapped onto the chosen COTS product set. Where gaps are identified, the requirements are structured to provide a more precise understanding that can be used to complete the Design. Design - The system is shaped and formed to meet all functional and supplemental requirements to enable the requirements to be developed into a working software system in the Implementation process.	• Software Component Design • Component Data Design • Component Behavior Design • Business Rules	Key focus on further exposing and understanding requirements, described as use cases, that are associated with new data structures and new data entry functionality. This leads to designing components associated with custom extensions that are related to new data structures and new entry functionality.	The Domain Model is further attributed to provide details about how and where the COTS-based model will be extended to support custom extensions being made to the COTS application to support identified gaps.The result is a Logical Database Design, which includes supporting business rules, facts, dimensions, etc.

		Design • User Interface Design • Design Specification • Logical Database Design		
Implementation	Through a number of iterative steps, the system is developed and unit tested.	• Conceptional and Functional Prototypes • Architectural Foundation • Development Environment and Implemented Database(s) • Software Components built, tested, and reviewed	Software components (for example, code, API's etc.) are built and unit-tested to satisfy the use cases.	The physical database objects are defined and created (for example, table, column names, constraints, etc).

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MANAGING ARTIFACT REVISIONS

Significant short-term cost savings may be realized by relaxing the standards for project documentation and lowering the level of ceremony for the project. The cost required to maintain consistency across various artifacts, as a project progresses through the implementation lifecycle, can be significant. Requirements artifacts do not necessarily need to be kept synchronized with the system design artifacts to successfully complete a project. However, the accuracy and availability of the final set of documentation may have a significant impact on the work required to manage or extend the production system. Since there is a cost associated with maintaining this consistency, it is essential to establish a concrete agreement related to that completeness and synchronization of the various project artifacts before providing a labor estimate for the project.

INFORMATION ARCHITECT ROLE

The information architect is a specialized system architect with expertise in Information (Data) Architecture. This role is sometimes referred to as a data architect.

In a Business Intelligence project, the information architect establishes the strategic data warehouse architecture and manages the integration of the increment with the data warehouse architecture. In order to accomplish this, they translate the future vision of the warehouse into the architecture which supports the extensibility and scaleability required for a data warehouse. This project role may develop or directly influence all strategies for the data warehouse specifically data acquisition and data quality, metadata, warehouse management, database design, data access, technical architecture and integration. The information architect provides input to more detailed

technical design efforts such as data acquisition, metadata, and data access to verify compatibility with the overall Information Architecture. The information architect outlines the data warehouse functions and how each is supported. The information architect works very closely with the technical architect to verify the physical layers of the architecture are consistent with, and fully support, the data warehouse vision for the enterprise. This synergism may extend all the way from scoping and planning the project through to the final architecture deliverables and customer review.

In addition, during a Business Intelligence project, the information architect examines the customer`s data sources; the extraction, transformation, and transportation (ETT) requirements; and the data models and database designs for the data warehouse. This project role works closely with the customer operations staff to obtain data source information including formats, definitions, currency, data interlinks, source system data models, and source system documentation. The information architect works with the business analyst to detail and categorize the requirements according to the data acquisition function, such as extraction, summarization, loading, etc. The information architect must understand the source data and the information requirements for the data warehouse, data mart, and metadata databases in order to map the source data to the target data structure. During this effort, the information architect works with the database designer to identify data shortfalls of the source data and identify data issues. This project role documents the error and exception handling requirements for data integrity and quality, determines the data acquisition tool requirements, and participates in the evaluation and selection of the tools. A information architect should be familiar with the scope of the increment and must verify that the data acquisition components support the information requirements. The information architect is responsible for working with the technical architect, data modeler, data access team, and data warehouse architect to verify the data acquisition process is an integrated component of the overall data warehouse solution.

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BUSINESS INTELLIGENCE (BI) SUPPLEMENTAL ACTIVITY GUIDELINES

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The Supplemental Activity Guidelines provided below are meant to assist teams in tailoring the workplan for a BI projects. They also provide some additional guidance for performing the OUM tasks and activities when implementing a Business Intelligence system that includes Enterprise Performance Management (EPM) or Business Intelligence Application (BI Apps) products. OUM defines an activity as "a set of tasks related either by topic, dependencies, data, common skills or work products."

Currently, guidance is provided for the following BI areas of focus:

- **BI Applications** (also referred to as BI Apps) - These engagements involve the delivery of projects utilizing commercial-off-the-shelf (COTS) business intelligence offerings such as EPM and other BI Applications. These projects also include some amount of custom work to extend the COTS application(s), as necessary. This work is often referred to as custom extensions.
- **Custom BI** (also referred to as Cust BI) - These engagements involve custom technological BI solutions that range from large-scale data warehouses to daily operational business intelligence applications, that are tailored to suit the specific needs of an enterprise. They do not include COTS offerings.

You should assume the guidance applies to all BI projects, including custom BI and BI Apps, unless the guidance explicitly refers to custom BI, BI Apps, and in many cases EPM.

Use the BI supplemental activity guidance provided on the left side of this table, in conjunction with the standard OUM activity overview on the right side of the table, in order to understand how BI impacts specific OUM activities.

Activity ID	Business Intelligence (BI) Supplemental Activity Guidelines	OUM Activity Overview (Links in this column refer to the OUM Activity Guidelines files)
Inception		
A.ACT.GRBI	Gather Business Requirements - Inception	Gather Business Requirements - Inception
A.ACT.ECBB	Establish Current Business Baseline	Establish Current Business Baseline
A.ACT.GR	Gather Solution Requirements	Gather Solution Requirements
A.ACT.CR	Consolidate Solution Requirements	Consolidate Solution Requirements
A.ACT.CCPI	Create Conceptual Prototype - Inception	Create Conceptual Prototype - Inception
A.ACT.GSP	Gather Supporting Requirements	Gather Supporting Requirements
A.ACT.SKSD	Specify Key Structure Definition	Specify Key Structure Definition

A.ACT.CMAHC	Create and Manage Ad Hoc Communications	Create and Manage Ad Hoc Communications
A.ACT.CEAW	Conduct Executive Alignment Workshop	Conduct Executive Alignment Workshop
A.ACT.TPT	Train Project Team	Train Project Team
A.ACT.CAWI	Conduct Alignment Workshops - Inception	Conduct Alignment Workshops - Inception
A.ACT.CCRA	Conduct Change Readiness Assessment	Conduct Change Readiness Assessment
A.ACT.DCMRCCI	Deploy Change Management Roadmap and Communication Campaign - Inception	Deploy Change Management Roadmap and Communication Campaign - Inception
A.ACT.LO	Lifecycle Objective Milestone	Lifecycle Objective Milestone
Elaboration		
B.ACT.DPS	Define Project Strategy	Define Project Strategy
B.ACT.DI	Define Infrastructure	Define Infrastructure
B.ACT.PEE	Prepare Environments - Elaboration	Prepare Environments - Elaboration
B.ACT.GBRE	Gather Business Requirements - Elaboration	Gather Business Requirements - Elaboration
B.ACT.DTP	Develop Test Plans	Develop Test Plans
B.ACT.SSC	Specify Software Configuration	Specify Software Configuration
B.ACT.PVC	Prototype and Validate Configuration	Prototype and Validate Configuration
B.ACT.PGA	Perform Gap Analysis	Perform Gap Analysis
B.ACT.DUCM	Develop Use Case Model	Develop Use Case Model
B.ACT.CCPE	Create Conceptual Prototype - Elaboration	Create Conceptual Prototype - Elaboration
B.ACT.CS	Consolidate Specification	Consolidate Specification
B.ACT.BSA	Baseline Software Architecture	Baseline Software Architecture
B.ACT.DUCD	Develop Use Case Details	Develop Use Case Details
B.ACT.AE	Analyze - Elaboration	Analyze - Elaboration
B.ACT.DE	Design - Elaboration	Design - Elaboration
B.ACT.DCCTSE	Develop Custom Component Test Scripts - Elaboration	Develop Custom Component Test Scripts - Elaboration
B.ACT.DSTSE	Develop System Test Scenarios - Elaboration	Develop System Test Scenarios - Elaboration
B.ACT.DVCSP	Develop and Validate Custom Software Prototypes	Develop and Validate Custom Software Prototypes
B.ACT.PCCTE	Perform Custom Component Testing - Elaboration	Perform Custom Component Testing - Elaboration
B.ACT.PSTE	Perform System Test - Elaboration	Perform System Test - Elaboration
B.ACT.PPM	Plan Performance Management	Plan Performance Management
B.ACT.PACDE	Prepare to Acquire and Convert Data - Elaboration	Prepare to Acquire and Convert Data - Elaboration
B.ACT.PBPII	Prepare Business Process Impact Inventory	Prepare Business Process Impact Inventory
B.ACT.DCMRCCE	Deploy Change Management Roadmap and Communication Campaign - Elaboration	Deploy Change Management Roadmap and Communication Campaign - Elaboration
B.ACT.LA	Lifecycle Architecture Milestone	Lifecycle Architecture Milestone
Construction		
C.ACT.FR	Finalize Requirements	Finalize Requirements

C.ACT.AC	Analyze - Construction	Analyze - Construction
C.ACT.DC	Design - Construction	Design - Construction
C.ACT.PTP	Perform Test Planning	Perform Test Planning
C.ACT.PEC	Prepare Environments - Construction	Prepare Environments - Construction
C.ACT.DCCTSC	Develop Custom Compoent Test Scripts - Construction	Develop Custom Compoent Test Scripts - Construction
C.ACT.DSTSC	Develop System Test Scenarios - Construction	Develop System Test Scenarios - Construction
C.ACT.DSITS	Develop Systems Integration Test Scenarios	Develop Systems Integration Test Scenarios
C.ACT.ICC	Implement Custom Components	Implement Custom Components
C.ACT.PCCTC	Perform Custom Component Testing - Construction	Perform Custom Component Testing - Construction
C.ACT.PST	Prepare for System Test	Prepare for System Test
C.ACT.PSTC	Perform System Test - Construction	Perform System Test - Construction
C.ACT.PSIT	Perform Systems Integration Test	Perform Systems Integration Test
C.ACT.PPT	Prepare for Performance Testing	Prepare for Performance Testing
C.ACT.PT	Prepare for Transition	Prepare for Transition
C.ACT.PCO	Prepare for Cutover	Prepare for Cut over
C.ACT.TI	Test Infrastructure	Test Infrastructure
C.ACT.PACDC	Prepare to Acquire and Convert Data - Construction	Prepare to Acquire and Convert Data - Construction
C.ACT.ACDC	Acquire and Convert Data - Construction	Acquire and Convert Data - Construction
C.ACT.PD	Produce Documentation	Produce Documentation
C.ACT.DCMRCCC	Deploy Change Management Roadmap and Communication Campaign - Construction	Deploy Change Management Roadmap and Communication Campaign - Construction
C.ACT.CJIA	Conduct Job Impact Analysis	Conduct Job Impact Analysis
C.ACT.CMAWC	Conduct Managers Alignment Workshop - Construction	Conduct Managers Alignment Workshop - Construction
C.ACT.DEUT	Design End-User Training	Design End-User Training
C.ACT.BEUT	Build End-User Training	Build End-User Training
C.ACT.TEUC	Train End Users - Construction	Train End Users - Construction
C.ACT.IOC	Initial Operational Capability Milestone	Initial Operational Capability Milestone
Transition		
D.ACT.SUAT	Support User Acceptance Test	Support User Acceptance Test
D.ACT.CPTST	Conduct Performance Test	Conduct Performance Test
D.ACT.CDT	Convert Data - Transition	Convert Data - Transition
D.ACT.PPE	Prepare Production Environment	Prepare Production Environment
D.ACT.CDT	Convert Data - Transition	Convert Data - Transition
D.ACT.DCMRCCT	Deploy Change Management Roadmap and Communication Campaign - Transition	Deploy Change Management Roadmap and Communication Campaign - Transition
D.ACT.CITA	Conduct IT Alignment	Conduct IT Alignment
D.ACT.TEUT	Train End Users - Transition	Train End Users - Transition

D.ACT.FD	Finalize Documentation	Finalize Documentation
D.ACT.GP	Go Production	Go Production
D.ACT.SP	System in Production Milestone	System in Production Milestone
Production		
E.ACT.MPSP	Manage Production System Performance	Manage Production System Performance
E.ACT.EPS	Evaluate Production System	Evaluate Production System
E.ACT.RPP	Resolve Production Problems	Resolve Production Problems
E.ACT.DCMRCCP	Deploy Change Management Roadmap and Communication Campaign - Production	Deploy Change Management Roadmap and Communication Campaign - Production
E.ACT.PF	Plan for Future	Plan for Future
E.ACT.DITTP	Deploy IT Transition Plan	Deploy IT Transition Plan
E.ACT.SO	Sign Off Milestone	Sign Off Milestone
Project Closure		
PC.ACT.GA	Gain Acceptance	Gain Acceptance
PC.ACT.CPC	Close Processes and Contract	Close Processes and Contract
PC.ACT.DLLAP	Document Lessons Learned and Archive Project	Document Lessons Learned and Archive Project

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INCEPTION

A.ACT.GBRI Gather Business Requirements - Inception

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

The business requirements for the BI solution are identified, refined, and prioritized as outlined in the OUM process.

It is essential that the Detail Business and System Objectives (RD.001) task be completed as a part of every project. This task defines the objectives for the BI project.

Business and/or system ***goals*** answers "why" the organization needs a particular requirement. Business and System ***Objectives*** answer "how" a requirement will help the business realize the goal. There may be multiple objectives that must be met in order for the business to realize its goal(s).

Common BI and EPM Business Goals:

- The solution must support improved business decision making.
- The data must represent "one version of the truth."
- The data (figures) must be consistent regardless of the view.
- The solution's structures must be adaptable to changing business needs.
- The data must be secure (making sensitive information accessible by only a named few).

Traceability of Business Objectives to Goals

An example of a many - to - one (business objectives to business goal) relationship:

Business Goal: The data must represent one version of the truth

Supporting Business Objectives:

- Merge the pertinent data from JDE, Hyperion, Oracle GL and OM
- Utilize a single source for the product master
- Apply consistent business rules and calculations for exchange rates

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
RD.001	BI Apps=>Rec. Custom BI=>Rec.	Detail Business and System Objectives	Business and System Objectives	It is possible that some initial form of these may be available as part of the project business case or bid materials or as part of OUM Envision materials that may have been developed when conceptualizing this project. If so, these objectives should be validated.
RD.005	BI Apps=>Rec. Custom BI=>Rec.	Create System Context Diagram	System Context Diagram	The System Context Diagram identifies the boundary for the system under development. It is a crucial element of scope management.
RD.011.1	BI Apps=>Opt. Custom BI=>Rec.	Develop Future Process Model	Future Process Model	BI and EPM projects may required the development of new business processes. For an EPM project, consider documenting the business processes affected by the implementation. For a BI Apps project, consider documenting any new or revised extract, transform, and load (ETL) processes.
RD.020	BI Apps=>Opt. Custom BI=>Rec.	Obtain High-Level Business Descriptions	High-Level Business Descriptions	This task involves identifying what the business does. For EPM or I Apps projects there may be limited value in documenting this information. Focus on obtaining business information related to gaps between requirements and functionality supported by EPM or BI Apps software.
RA.015	BI Apps=>Opt. Custom BI=>Rec.	Develop Business Use Case Model	Business Use Case Model	
RD.012	BI Apps=>Opt. Custom BI=>Rec.	Document Present and Future Organizational Structures	Present and Future Organizational Structures	Consider this task when it is necessary to understand the organizational structures that will have an effect on the configuration or functionality of the system. You should strictly limit your examination to only those structures that are relevant to the system being implemented.
RD.015	BI Apps=>Opt. Custom BI=>Rec.	Determine KPI Collection and Reporting Strategy	Key Business Strategies and Objectives	Most businesses that are implementing BI and EPM systems already know the KPI's they want to measure or improve. This information may have been provided as part of the project's business case. Frequently, the BI or EPM system is intended to become a primary pillar in the collection and reporting strategy

				for the enterprise.
RD.065	BI Apps=>Rec. Custom BI=>Rec.	Develop Domain Model	Domain Model	A comprehensive data model exists for EPM and BI Apps products. A Domain Model is used to establish data requirements for custom developed BI systems or to capture new data requirements for custom extensions to EPM and BI Apps products.

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A.ACT.ECBB Establish Current Business Baseline

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Optional

The RD.030 task is recommended if examination of the current business processes is needed.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
RD.030	BI Apps=>Opt. Custom BI=>Opt.	Develop Current Business Process Model	Current Business Process Model	Consider executing this task when the project scope includes a business process improvement initiative or when it is explicitly necessary to understand existing processes. It is more likely applicable for EPM projects that are seeking to improved business processes around enterprise performance management.
RD.034	BI Apps=>Opt. Custom BI=>Opt.	Document Current Business Baseline Metrics	Current Business Baseline Metrics	BI / EPM projects are likely to be initiated by a business in order to baseline and improve specific KPI's. Examine the project startup materials to confirm the status of KPI baselines.

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A.ACT.GR Gather Solution Requirements

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended

Custom BI	Recommended

The Develop Glossary (RD.042) task is highly recommended for all Custom BI, EPM, and BI Apps projects. Terms that are used to classify or identify data should be part of the project's glossary. Terms that will be used across business silos or by both the IT organization and the project team should be reviewed and documented.

The project glossary is particularly useful on BI Apps projects, as the meaning of common business terms can be an effective method for eliciting business rules, identifying missing, or redundant data elements and gaining agreement on calculations.

A Use Case Model (RA.023) should be created for EPM and BI Apps projects that require customizations or enhancements to the EPM or BI Apps product software. It is not necessary to model or specify existing product functionality unless doing so adds measurable value. The focus of the effort in this activity should be on modeling use cases to document gaps between the business requirements and the functionality provided by the chosen product set.

For BI Apps projects, there may be little emphasis on use cases at the black box level. Users are keying in data that is used by the system. The exception to this might be for identifying report requirements for displaying the captured data.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
RD.042.1	BI Apps=>Rec. Custom BI=>Rec.	Develop Glossary	Glossary	Document acronyms and organization or project specific terms. This task should be considered completed with a standard glossary unless new terms exist.
RD.045.1	BI Apps=>Rec. Custom BI=>Rec.	Prioritize Requirements (MoSCoW)	MoSCoW List	
RD.055	BI Apps=>Rec. Custom BI=>Rec.	Detail Supplemental Requirements	Supplemental Requirements	This task covers non-functional requirements.
RD.130.1	BI Apps=>Rec. Custom BI=>Rec.	Develop Baseline Architecture Description	Architecture Description	
RA.023.1	BI Apps=>Rec. Custom BI=>Rec.	Develop Use Case Model	Use Case Model	Use cases that must be developed in order to fill gaps in COTS functionality.
RA.027.1	BI Apps=>Rec. Custom BI=>Rec.	Identify Candidate Business Rules	Candidate Business Rules	Use the MoSCoW List (RD.045) to support traceability from use cases to business rules and vise versa.
RA.028.1	BI Apps=>Opt. Custom BI=>Rec.	Populate Business Rules Repository	Populated Business Rules Repository	Business rules may be included in the ETL tool or metadata tool.

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A.ACT.CR Consolidate Solution Requirements

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

The Requirements Specification (RD.140) task is provided to allow project teams to consolidates a number of requirements-related work products into a single document to facilitate a review of a consolidated requirements specification and for use as a deliverable to satisfy a project milestone. As shown in the RD.140 task guideline and template you might include all relevant requirements artifacts within the Requirements Specification work product, or simply provide links to the various requirements-related artifacts.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
RD.140.1	BI Apps=>Rec. Custom BI=>Rec.	Create Requirements Specification	Requirements Specification	
RD.150.1	BI Apps=>Rec. Custom BI=>Rec.	Review Requirements Specification	Results Reviewed	

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A.ACT.CCPI Create Conceptual Prototype - Inception

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

In this activity, you create the Conceptual Prototype. The purpose of this low fidelity prototype is to show the users your understanding of an agreed on set of requirements. In reality, business users may not be able to make design decisions without actually seeing dashboards and actual reports. If this prototype is used, the following tasks should be considered for a BI project.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
IM.005.1	BI Apps=>Rec. Custom BI=>Rec.	Develop Conceptual Prototype	Conceptual Prototype	

RA.030.1	BI Apps=>Rec. Custom BI=>Rec.	Validate Conceptual Prototype	Validated Conceptual Prototype	Archiving Meeting Minutes of a Review Session is an acceptable form of validation.

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A.ACT.GSP Gather Supporting Requirements

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

This activity includes the gathering of supporting requirements that are not directly related to the business and system objectives. For BI projects you would typically complete the recommended tasks. The optional tasks depend on the requirements for the project.

While OUM differentiates between different types of requirements - functional, supplemental, etc. - the business representatives will frequently provide supporting requirements as a natural part of requirements gathering. This includes documenting constraints. You should be prepared to document all types of requirements even when the focus of a workshop is on functional requirements.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
CV.010	BI Apps=>Rec. Custom BI=>Rec.	Define Data Acquisition and Conversion Requirements	Data Acquisition and Conversion Requirements	Use this task to outline requirements for the initial and ongoing data loads. This includes data enrichment or data supplementation from from third party providers (e.g. currency exchange rates needed for data transformation).
TE.005.1	BI Apps=>Rec. Custom BI=>Rec.	Determine Testing Requirements	Testing Requirements	Use this task to evaluate the high-level strategy and requirements for testing the implemented system.
PT.010	BI Apps=>Opt. Custom BI=>Opt.	Conduct Performance Management Workshop	Performance Management Workshop Results	The Performance Management Workshop is important for projects that have contractually delegated the bulk of responsibility for Performance Management and Performance Testing to the customer or third-party provider.
TA.010	BI Apps=>Rec. Custom BI=>Rec.	Conduct Technical Architecture Workshop	Technical Architecture Workshop Results	Infrastructure information and system constraints need to be collected from the technical support team.
DO.010	BI Apps=>Rec. Custom BI=>Rec.	Define Documentation Requirements and Strategy	Documentation Requirements and Strategy	

RD.070	BI Apps=>Rec. Custom BI=>Opt.	Determine Audit and Control Requirements	Audit and Control Requirements	Use this task to identify the high level requirements and policies that affect business and system security, control and audit procedures.

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A.ACD.SKSD Specify Key Structure Definition

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Not Applicable

This activity includes the design of key business architecture structures, which have an impact across the entire application system. It also includes the definition of the data/values needed to configure those key business architecture structures.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
MC.010.1	BI Apps=>Rec. Custom BI=>N/A	Define Business Data Structures	Business Data Structures	Identify needed dimensions and/or facts. May use domain modeling, ER diagrams and/or Dimensional Stars.
MC.020	BI Apps=>Rec. Custom BI=>N/A	Define Business Data Structure Setups	Business Data Structure Setups	Data modeling that results from this task can be done as either an entity or class model or the as a traditional star diagram. It is recommended that only the structures that support additional or extended dimensions/facts be modeled. It is not necessary to identify/model COTS structures.

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A.ACT.CMAHC Create and Manage Ad Hoc Communication

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Optional

This activity is part of OUM's comprehensive set of Organizational Change Management materials that help to prepare the organization for the adoption of the new

system. This work is frequently the responsibility of the customer and should always be scaled to suit the needs of the project.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
OCM.010	BI Apps=>Opt. Custom BI=>Opt.	Create and Manage Ad Hoc Communications	Ad Hoc Communications	

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A.ACT.CEAW Conduct Executive Alignment Workshop

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Optional

This activity is part of OUM's comprehensive set of Organizational Change Management materials that help to prepare the organization for the adoption of the new system. This work is frequently the responsibility of the customer and should always be scaled to suit the needs of the project.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
OCM.020	BI Apps=>Opt. Custom BI=>Opt.	Prepare for Executive Alignment Workshop	Executive Alignment Workshop Materials	
OCM.030	BI Apps=>Opt. Custom BI=>Opt.	Conduct Executive Alignment Workshop	Executive Execution Plan / Governance Rules	Usually a steering committee helps make decisions.
OCM.040	BI Apps=>Opt. Custom BI=>Opt.	Build and Deploy Sponsorship Program	Sponsorship Program	

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A.ACT.TPT Train Project Team

	Activity

BI Area of Focus	Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

This activity can be used to prepare a learning plan for the project team, as well as establishing the learning environment and lastly, put the plan into action by conducting the team learning events. Many EPM and BI Apps customers may have an internal team of developers who need training before configuration and customization commences. Review the contract to identify whether Oracle or the customer is responsible for this activity.

Although most tasks here are listed as optional, it is recommended that they are taken into consideration during the early stages of the project.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
TR.010.1	BI Apps=>Rec. Custom BI=>Rec.	Define Training Strategy	Education and Training Strategy	Users of a new EPM and BI Apps application will need training. Consider adding the Glossary to the training material.
TR.020	BI Apps=>Rec. Custom BI=>Rec.	Prepare Project Team Learning Plan	Project Team Learning Plan	
TR.030	BI Apps=>Rec. Custom BI=>Rec.	Prepare Project Team Learning Environment	Working Learning Environment	
TR.040	BI Apps=>Rec. Custom BI=>Rec.	Develop Project Team Learningware	Project Team Learningware	
TR.050	BI Apps=>Rec. Custom BI=>Rec.	Conduct Project Team Learning Events	Project Team Learning Events Administration	

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A.ACT.CAWI Conduct Alignment Workshops - Inception

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Optional

This activity is part of OUM's comprehensive set of Organizational Change Management materials that help to prepare the organization for the adoption of the new system. This work is frequently the responsibility of the customer and should always be scaled to suit the needs of the project.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
OCM.050	BI Apps=>Opt. Custom BI=>Opt.	Prepare for Team-Building Workshop	Team-Building Workshop Materials	
OCM.060	BI Apps=>Opt. Custom BI=>Opt.	Conduct Team-Building Workshop	Cohesive Project Team	
OCM.070	BI Apps=>Opt. Custom BI=>Opt.	Design Managers Project Alignment Workshop	Managers Project Alignment Workshop Materials	
OCM.080	BI Apps=>Opt. Custom BI=>Opt.	Conduct Managers Project Alignment Workshop	Aligned Managers	
OCM.090	BI Apps=>Opt. Custom BI=>Opt.	Design Change Agent Workshop	Change Agent Workshop Materials	
OCM.100	BI Apps=>Opt. Custom BI=>Opt.	Conduct Change Agent Workshop	Enabled Change Agents	

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A.ACT.CCRA Conduct Change Readiness Assessment

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Optional

This activity is part of OUM's comprehensive set of Organizational Change Management materials that help to prepare the organization for the adoption of the new system. This work is frequently the responsibility of the customer and should always be scaled to suit the needs of the project.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
OCM.110	BI Apps=>Opt. Custom BI=>Opt.	Develop Change Readiness Assessment Strategy and Tools	Change Readiness Assessment Strategy and Tools	

OCM.120	BI Apps=>Opt. Custom BI=>Opt.	Conduct Change Readiness Assessment and Analyze Data	Change Readiness Assessment Analysis and Recommendations	
OCM.130	BI Apps=>Opt. Custom BI=>Opt.	Build Communication Strategy and Change Management Roadmap	Communication Strategy Change Management Roadmap	

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A.ACT.DCMRCCI Deploy Change Management Roadmap and Communication Campaign - Inception

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Optional

This activity is part of OUM's comprehensive set of Organizational Change Management materials that help to prepare the organization for the adoption of the new system. This work is frequently the responsibility of the customer and should always be scaled to suit the needs of the project.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
OCM.140	BI Apps=>Opt. Custom BI=>Opt.	Develop Communication Campaign	Communication Campaign	
OCM.150.1	BI Apps=>Opt. Custom BI=>Opt.	Conduct Change Management Roadmap and Communication Campaign	Change Management Roadmap and Communication Events	

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A.ACT.LO Lifecycle Objective Milestone

For BI projects, the following recommended activities should be completed in order to achieve this milestone in the project:

Activity ID	Activity Name	BI Applications (including EPM)	Custom BI
A.ACT.GBRI	Gather Business Requirements - Inception	Recommended	Recommended
A.ACT.ECBB	Establish Current Business Baseline	Optional	Optional

A.ACT.GR	Gather Solution Requirements	Recommended	Recommended
A.ACT.CR	Consolidate Solution Requirements	Recommended	Recommended
A.ACT.CCPI	Create Conceptual Prototype - Inception	Recommended	Recommended
A.ACT.GSP	Gather Supporting Requirements	Recommended	Recommended
A.ACT.SKSD	Specify Key Structure Definition	Recommended	Not Applicable
A.ACT.CMAHC	Create and Manage Ad Hoc Communications	Optional	Optional
A.ACT.CEAW	Conduct Executive Alignment Workshop	Optional	Optional
A.ACT.TPT	Train Project Team	Recommended	Recommended
A.ACT.CAWI	Conduct Alignment Workshops - Inception	Optional	Optional
A.ACT.CORA	Conduct Organizational Readiness Assessment	Optional	Optional
A.ACT.DCMRCCI	Deploy Change Management Roadmap / Communication Campaign - Inception	Optional	Optional

Note: When the optional activities listed above are applicable to your BI project, it is recommended that you complete them in order to achieve this milestone.

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ELABORATION

B.ACT.DPS Define Project Strategy

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

The BI and EPM View objective for this activity is to develop project strategies for application extension, Testing, Data Acquisition and Conversion, Documentation, and Transition of the solution. Some tasks have been marked as optional; however, it is recommended that each task is reviewed to confirm if it is applicable.

Note: It is reasonable, especially on smaller projects, for these project strategy artifacts to be incorporated into or referenced by the Project Management Plan (PMP).

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
DS.020	BI Apps=>Rec. Custom BI=>N/A	Define Application Extension Strategy	Application Extension Strategy	
TE.005.2	BI Apps=>Rec. Custom BI=>Rec.	Determine Testing Requirements	Testing Requirements	
TE.010	BI Apps=>Rec.	Develop Testing Strategy	Testing Strategy	

	Custom BI=>Rec.			
CV.020	BI Apps=>Rec. Custom BI=>Rec.	Define Data Acquisition, Conversion and Data Quality Strategy	Data Acquisition, Conversion, and Data Quality Strategy	
DO.020	BI Apps=>Opt. Custom BI=>Rec.	Define Documentation Standards and Procedures	Documentation Standards and Procedures	Documentation Standards and Procedures (DO.020) for EPM - BI Apps projects largely depends on project requirements and contractual obligations.
TR.010.2	BI Apps=>Opt. Custom BI=>Rec.	Define Training Strategy	Education and Training Strategy	
TS.020.1	BI Apps=>Rec. Custom BI=>Rec.	Define Cutover Strategy	Cutover Strategy	

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B.ACT.DI Define Infrastructure

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

The Define Infrastructure activity consists of producing the requirements for the Technical Architecture infrastructure for the EPM - BI Apps system. All tasks in this activity are recommended when implementing a EPM - BI Apps system.

Note: You might consider consolidating all of the Technical Architecture work products into one or more consolidated artifacts or deliverables, as appropriate to the structure of the project.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
TA.020	BI Apps=>Rec. Custom BI=>Rec.	Define Technical Architecture Requirements and Strategy	Technical Architecture Requirements and Strategy	
TA.030	BI Apps=>Rec. Custom BI=>Rec.	Define Integration Requirements and Strategy	Integration Architecture Strategy	
TA.040	BI Apps=>Rec. Custom BI=>Rec.	Define Reporting and Information Access Strategy	Reporting and Information Access Strategy	

TA.050	BI Apps=>Rec. Custom BI=>Rec.	Define Disaster Recovery Strategy	Disaster Recover Strategy	
TA.060	BI Apps=>Rec. Custom BI=>Rec.	Define System Operations and Management Strategy	System Operations and Management Strategy	
TA.070	BI Apps=>Rec. Custom BI=>Rec.	Define Initial Architecture and Application Mapping	Initial Architecture and Application Mapping	
TA.080	BI Apps=>Rec. Custom BI=>Rec.	Define Backup and Recovery Strategy	Backup and Recovery Strategy	
TA.090	BI Apps=>Rec. Custom BI=>Rec.	Develop Security and Control Strategy	Security and Control Strategy	

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B.ACT.PEE Prepare Environments - Elaboration

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

In this activity, you create the Development Environment that you will need during the development stage of a BI or EPM implementation project. You should begin planning for the other environments that must be created to support QA and production.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
IM.007.1	BI Apps=>Rec. Custom BI=>Rec.	Prepare Development Environment	Development Environment	The Development Environment along with database and other critical items must be in place for software configuration and custom extensions to be developed in the Implement Components (IM.050) task.
TE.018.1	BI Apps=>Opt. Custom BI=>Rec.	Prepare Static Test Data	Static Test Data	You may be using the customer's source Development Environment for input data. This may be nothing more than a check-off item that data is ready for testing. It is important that the test data does have necessary characteristics/attributes in order to test the acceptance criteria test conditions established in the Acceptance Test Plan (TE.082)
TE.038.1	BI Apps=>Opt.	Prepare Integration Test	Integration Test Environment	The Integration Test Environment typically matches the

	Custom BI=>Rec.	Environment		Development Environment. In many cases there is only one test environment. A test environment including database(s), and other critical items, must be in place for software configuration and custom extensions to be developed in the Implement Components (IM.050) task.
TE.060.1	BI Apps=>Rec. Custom BI=>Rec.	Prepare System Test Environment	System Test Environment	The Integration Test Environment typically matches the Development Environment. In many cases there is only one test environment. A test environment including database(s), and other critical items, must be in place for software configuration and custom extensions to be developed in the Implement Components (IM.050) task.
CV.030.1	BI Apps=>Opt. Custom BI=>Rec.	Prepare Conversion Environment (Initial Load)	Conversion Environment (Initial Load)	The Conversion Environment typically matches the Development Environment. In many cases there is only one test environment. A test environment including database(s), and other critical items, must be in place for software configuration and custom extensions to be developed in the Implement Components (IM.050) task.
DO.040	BI Apps=>Rec. Custom BI=>Rec.	Prepare Documentation Environment	Documentation Environment	A Documentation Environment needs to be established in order to achieve the objective of completing the documentation to the agreed to levels. This environment could be as simple as a shared directory on a client server, or preferably, a collaborative knowledge area.

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B.ACT.GBRE Gather Business Requirements - Elaboration

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

The Future Process Model (RD.011.2) includes process flow diagrams of the events and business processes that the applications and the associated functions of the business area will support with regards to BI and EPM projects. In some cases, business processes will be affected by the introduction of the EPM and BI Apps software functionality.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
RD.011.2	BI Apps=>Opt. Custom BI=>Rec.	Develop Future Process Model	Future Process Model	BI and EPM projects may required the development of new business processes. For an EPM project, consider documenting the business process affected by the implementation. For a BI Apps project, consider documenting any new or revised extract,

				transform, and load (ETL) processes.
RA.055.1	BI Apps=>Rec. Custom BI=>N/A	Document Business Procedures	Business Procedure Documentation	This information may be needed to support new user training once the system has been implemented.
MC.030	BI Apps=>Rec. Custom BI=>N/A	Map Business Requirements	Business Requirements Mapping Form	

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B.ACT.DTP Develop Test Plans

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

The BI and EPM View objective for this activity is to develop the test plans. During the BI and EPM project, testing occurs throughout development or construction. It is important to keep a formal testing results log throughout the testing cycles.

Note: It is reasonable, where appropriate, to combine test plans into a single work product.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
TE.015.1	BI Apps=>Opt. Custom BI=>Rec.	Develop Integration Test Plan	Integration Test Plan	The scope of testing may differ for a custom BI versus EPM versus a BI Apps project.
TE.050.1	BI Apps=>Rec. Custom BI=>Rec.	Develop System Test Plan	System Test Plan	The scope of testing may differ for a custom BI versus EPM versus a BI Apps project.
TE.080	BI Apps=>Opt. Custom BI=>Rec.	Develop Systems Integration Test Plan	Systems Integration Test Plan	
TE.082	BI Apps=>Rec. Custom BI=>Rec.	Develop Acceptance Test Plan	Acceptance Test Plan	Acceptance criteria defined in the Acceptance Test Plan is very important, as it should minimize ambiguities later on when testing actually occurs as it clearly defines what was agreed upon by the customer.

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B.ACT.SSC Specify Software Configuration

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Not Applicable

This activity allows for the definition and documentation of all setup information, including business data structures, rules, calculations, permissions, etc., not previously addressed in the Specify Key Structure Definition activity.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
MC.010.2	BI Apps=>Rec. Custom BI=>N/A	Define Business Data Structures	Business Data Structures	
MC.040	BI Apps=>Rec. Custom BI=>N/A	Gather Setup Information	Setup Information	
MC.050.1	BI Apps=>Rec. Custom BI=>N/A	Define Application Setups	Application Setup Documents	
MC.060	BI Apps=>Rec. Custom BI=>N/A	Document Functional Security	Functional Security Setup Information	

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B.ACT.PVC Prototype and Validate Configuration

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Not Applicable

This activity is used to validate the standard application functionality. It also includes validation of the standard reports and forms, provided with the BI Apps (for example, EPM) system functionality being implemented.

	Task Consideration	Task Name		

Task ID	(Link to BI Supp Task Guidance)	(Link to OUM Base Task Overview)	Work Product Name	Task Notes
TE.025.1	BI Apps=>Rec. Custom BI=>N/A	Create System Test Scenarios	System Test Scenarios	
MC.070	BI Apps=>Rec. Custom BI=>N/A	Prepare Configuration Prototype Environment	Configuration Prototype Environment	
MC.080	BI Apps=>Rec. Custom BI=>N/A	Conduct Configuration Prototyping Workshop	Validated Configuration	During this workshop, you validate that the packaged application functionality, as configured based on the results of the Specify Software Configuration activity, meets the documented requirements.
MC.090.2	BI Apps=>Opt. Custom BI=>N/A	Conduct Reporting Fit Analysis	Reporting Fit Analysis	

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B.ACT.PGA Perform Gap Analysis

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Not Applicable

Within this activity the project team reviews any proposed business process changes and other modifications (setups, interfaces, custom components, etc.) for gaps previously identified. Alternative ways to address open requirements are determined and estimates for each alternative prepared. As a final step the alternative solutions are reviewed with the customer and the best alternative is determined based on the customer's preference. It is important to remember that possibly not all gaps will be considered for the initial release - Contract obligations, budget, and priorities all impact the implementation plan.

This activity should be completed if prototypes have been developed and validated.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
MC.100	BI Apps=>Rec. Custom BI=>N/A	Define and Estimate Application Extensions	Application Extension Definition and Estimates	
MC.110	BI Apps=>Rec. Custom BI=>N/A	Define Gap Resolutions	Gap Resolutions	

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B.ACT.DUCM Develop Use Case Model

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

In this activity, the Use Case Model (RA.023) should be created for all BI and EPM projects that require customization, or enhancement to the system. It is recommended that the Use Case Model be created at the same time as the MoSCoW List as they are very closely linked. The Use Case Model shows a structured representation of the project use cases/requirements.

During Elaboration, all specified use cases should be reviewed and the appropriate updates made to the Use Case Model (see also RA.180 Review Use Case Model).

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
RA.023.2	BI Apps=>Rec. Custom BI=>Rec.	Develop Use Case Model	Use Case Model	

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B.ACT.CCPE Create Conceptual Prototype - Elaboration

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

In this activity, you review and modify the Conceptual Prototype. The purpose of this prototype is to show the users your understanding of an agreed on set of requirements. Reviews may also verify that the flow of data is as expected.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
IM.005.2	BI Apps=>Rec.	Develop Conceptual Prototype	Develop Conceptual Prototype	This may be a Proof of Concept (POC) or a limited functional

	Custom BI=>Rec.			prototype. Exposing the customer early in the project to the prototype helps set expectations.
RA.030.2	BI Apps=>Rec. Custom BI=>Rec.	Validate Conceptual Prototype	Validated Conceptual Prototype	Create minutes to document changes or use the template.

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B.ACT.CS Consolidate Specification

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

The objective for this activity is to refine the software requirements and review the Use Case Model and Use Case Specifications and prepare for detail analysis and design of the BI / EPM solution.

Please ensure that either the Review Use Case Model (RA.180.1) task is completed or additional columns are added to the MoSCoW List in order to ensure that the Use Case Model is accepted by all parties.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
RD.042.2	BI Apps=>Rec. Custom BI=>Rec.	Develop Glossary	Glossary	It is essential that a project glossary is developed and reviewed as it avoids misinterpretation of EPM - BI Apps and business terms. The Develop Glossary (RD.042) task is highly recommended for all EPM / BI Apps projects.
RD.045.2	BI Apps=>Rec. Custom BI=>Rec.	Prioritize Requirements (MosCoW)	MoSCoW List	The MoSCoW List (RD.045) is one of the most important work products for any project. This task is used to review and modify the MoSCoW List, as appropriate.
RD.140.2	BI Apps=>Rec. Custom BI=>Rec.	Create Requirements Specification	Requirements Specification	
RD.150.2	BI Apps=>Rec. Custom BI=>Rec.	Review Requirements Specification	Results Reviewed	A review workshop may support cross-functional acceptance of the requirements. Meeting minutes should be archived and any updates to functionality should be appended to the Requirements Specification (RD.140). Typically, review results are captured in detailed format within meeting minutes.
RA.027.2	BI Apps=>Rec. Custom BI=>Rec.	Identify Candidate Business Rules	Candidate Business Rules	The task, Identify Candidate Rules (RA.027.2) highlights business rules as statements that describe business policies that constrain data and/or functionality.

RA.028.2	BI Apps=>Opt. Custom BI=>Rec.	Populate Business Rules Repository	Populated Business Rules Repository	Business rules may be included in the ETL tool or metadata tool.
RA.180.1	BI Apps=>Rec. Custom BI=>Rec.	Review Use Case Model	Results Reviewed	A review workshop may support cross-functional acceptance of the requirements. Meeting minutes should be archived and any updates to functionality may be appended to the Requirements Specification (RD.140).

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B.ACT.BSA Baseline Software Architecture

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

This activity is used to further develop the Architecture Description (RA.035) for the BI and EPM solution.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
RA.035	BI Apps=>Rec. Custom BI=>Rec.	Develop High-Level Software Architecture Description	Architecture Description	
AN.040.1	BI Apps=>Opt. Custom BI=>Rec.	Develop Analysis Architecture Description	Architecture Description	
DS.040.1	BI Apps=>Opt. Custom BI=>Rec.	Develop Design Architecture Description	Architecture Description	

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B.ACT.DUCD Develop Use Case Details

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended

Custom BI	Recommended

For BI Apps and EPM, the Use Case Specification may be needed for use cases defined as a 'Gap' for BI Apps and EPM projects.

It is important to understand the audience/recipient of the use case specification. Fitness for purpose should be considered when deciding on what use cases should be modeled or detailed. For BI Apps projects, use cases are often at the white-box level (knowledge of software internal workings). An alternative to detailing out a white box use case may be to create an installation or configuration guide or to develop a system sequence diagram or component diagram. Review the sample documentation before making this decision.

During Elaboration, all specified use cases should be reviewed and the appropriate updates made to the Use Case Model (see also RA.180 Review Use Case Model).

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
RA.024.1	BI Apps=>Rec. Custom BI=>Rec.	Develop Use Case Details	Use Case Specification	

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B.ACT.AE Analyze - Elaboration

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

This activity allows for analyzing the captured requirements by refining and structuring them via various conceptual object models, ultimately organized into the Analysis Model, which, in turn, leads to the Design Model.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
AN.050.1	BI Apps=>Opt. Custom BI=>Rec.	Analyze Data	Data Analysis	
AN.060.1	BI Apps=>Rec. Custom BI=>Rec.	Analyze Behavior	Behavior Analysis	
AN.070.1	BI Apps=>Rec. Custom	Analyze Business Rules	Business Rules Analysis	

	BI=>Rec.			
AN.090.1	BI Apps=>Rec. Custom BI=>Rec.	Analyze User Interface	User Interface Analysis	
AN.100.1	BI Apps=>Opt. Custom BI=>Rec.	Prepare Analysis Specification	Analysis Specification	If necessary, the Analysis Specification (AN.100) task can be used to create a single document that consolidates the work products from the tasks in this activity. It would also be reasonable to document the results of analysis and design into the design artifacts and not product separate analysis artifacts.
AN.110.1	BI Apps=>Opt. Custom BI=>Rec.	Review Analysis Model	Reviewed Analysis Model	Meeting minutes may be used to document results of the review process.

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B.ACT.DE Design - Elaboration

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

This activity allows for shaping the system to meet all functional and supplemental requirements.

The optional tasks are generally performed when there is a need to gain further clarification of the business requirements represented in the use cases. This typically applies to those requirements that can only be satisfied by custom-built components, which extend the functionality of the EPM or the BI Apps system and/or integrate the EPM or BI Apps system with another system or legacy systems.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
DS.050	BI Apps=>Rec. Custom BI=>Rec.	Determine Design and Build Standards	Design and Build Standards	Oracle standards should be utilized, if customer-specific standards do not exist, or are not sufficient to meet needs. This is typically an item reviewed for existence and compliance if a project evaluation is performed.
DS.060	BI Apps=>Rec. Custom BI=>Rec.	Define Business Rules Implementation Strategy	Business Rules Implementation Strategy	Candidate rules should already have been originally captured and should be reviewed during the definition of the Business Rules Implementation Strategy.
DS.080.1	BI Apps=>Rec. Custom BI=>Rec.	Design Software Components	Software Component Design	
DS.090.1	BI Apps=>Rec.	Design Data	Component Data Design	

	Custom BI=>Rec.			
DS.100.1	BI Apps=>Rec. Custom BI=>Rec.	Design Behavior	Component Behavior Design	
DS.110.1	BI Apps=>Rec. Custom BI=>Rec.	Design Business Rules	Business Rules Design	
DS.130.1	BI Apps=>Rec. Custom BI=>Rec.	Design User Interface	User Interface Design	
DS.140.1	BI Apps=>Rec. Custom BI=>Rec.	Prepare Design Specification	Design Specification	
DS.150.1	BI Apps=>Rec. Custom BI=>Rec.	Develop Database Design	Logical Database Design	For a custom BI development project, this Logical Database Design for the entire system would be produced in this task. For a COTS implementation with extensions, this design would likely be limited to the logical database design for extensions to the COTS database.
DS.160.1	BI Apps=>Rec. Custom BI=>Rec.	Review Design Model	Reviewed Design Model	Meeting minutes from review sessions of the Design Specification (DS.140) or in the Design Specification (DS.140) can serve as acceptance. Hard-copy sign-off can be added to the Project Notebook (if one is kept).

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B.ACT.DCCTSE Develop Custom Component Test Scripts - Elaboration

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Recommended

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
TE.020.1	BI Apps=>Opt. Custom BI=>Rec.	Develop Unit Test Scripts	Unit Test Scripts	

TE.035.1	BI Apps=>Opt. Custom BI=>Rec.	Create Integration Test Scenarios	Integration Test Scenarios	

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B.ACT.DSTSE Develop System Test Scenarios - Elaboration

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Recommended

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
TE.025.2	BI Apps=>Opt. Custom BI=>Rec.	Create System Test Scenarios	System Test Scenarios	

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B.ACT.DVCSP Develop and Validate Custom Software Prototypes

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

This activity supports the development and validation of various prototypes of custom components for the BI and EPM system and performing a component review.

Validation of functional prototypes for custom extensions is generally limited to custom components that warrant development of a prototype at this stage of the project due to the technical complexity, or other risks associated with them.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes

IM.010	BI Apps=>Rec. Custom BI=>Rec.	Develop Functional Prototype	Functional Prototype	
RA.085	BI Apps=>Rec. Custom BI=>Rec.	Validate Functional Prototype	Validated Functional Prototype	
IM.020	BI Apps=>Opt. Custom BI=>Rec.	Develop Architectural Foundation	Architectural Foundation	
IM.085	BI Apps=>Opt. Custom BI=>Rec.	Develop User Interface Standards Prototype	User Interface Standards Prototype	The User Interface Standards Prototype (IM.085) is used to validate revisions that might be required to be made to the out-of-the-box application system user interface in order to meet the implementing organization's requirements. This is generally applicable to those EPM and BI Apps systems that require tailoring of the standard user interface.
RA.095	BI Apps=>Opt. Custom BI=>Rec.	Validate User Interface Standards Prototype	Validated User Interface Standards Prototype	
IM.040.1	BI Apps=>Opt. Custom BI=>Rec.	Implement Database	Physical Database Design	
IM.055.1	BI Apps=>Opt. Custom BI=>Rec.	Perform Rules Business Implementation (Rules Engine)	Implemented Business Rules (Rules Engine)	
IM.060.1	BI Apps=>Opt. Custom BI=>Rec.	Perform Component Review	Results Reviewed	Document the review results in the minutes, which are added to the finalized documentation.

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B.ACT.PCCTE Perform Custom Component Testing - Elaboration

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Recommended

Task ID	Task Consideration (Link to BI Supp Task	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes

	Guidance)			
TE.030.1	BI Apps=>Opt. Custom BI=>Rec.	Perform Unit Test	Unit -Tested Components	
TE.040.1	BI Apps=>Opt. Custom BI=>Rec.	Perform Integration Test	Integration-Tested Components	

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B.ACT.PSTE Perform System Test - Elaboration

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Recommended

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
TE.070.1	BI Apps=>Opt. Custom BI=>Rec.	Perform System Test	System-Tested Applications	

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B.ACT.PPM Plan Performance Management

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Optional

This activity consists of producing the requirements, strategy, testing models, scenarios, test scripts, and program designs for Performance Management and Testing.

Note: There are always some level of performance and tuning expectations. However this optional activity, and associated set of tasks are strongly recommended if performance requirements are spelled out in the contract.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
PT.020	BI Apps=>Opt. Custom BI=>Opt.	Define Performance Management Requirements and Strategy	Performance Management Requirements and Strategy	
PT.030	BI Apps=>Opt. Custom BI=>Opt.	Identify Performance Testing Strategy	Performance Testing Strategy	
PT.040	BI Apps=>Opt. Custom BI=>Opt.	Identify Performance Testing Models and Scenarios	Performance Testing Models and Scenarios	
PT.050	BI Apps=>Opt. Custom BI=>Opt.	Design Performance Test Scripts and Programs	Performance Test Scripts and Program(s) Design	
PT.060	BI Apps=>Opt. Custom BI=>Opt.	Design Performance Test data and Load Programs	Designed Performance Test Data and Designed Test Load Programs	

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B.ACT.PACDE Prepare to Acquire and Convert Data - Elaboration

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

In an BI and EPM implementation project, this activity consists of producing the standards and completing the high-level data mapping needed to confirm that the application software supports the customer organization's business processes and legacy data requirements. Conversion component design and development tasks are executed in a similarly named activity in the Construction Phase.

For BI and EPM implementation projects, it is recommended that an adequate Security Model has been developed before starting the data mapping task.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
CV.025	BI Apps=>Rec. Custom BI=>Rec.	Define Data Acquisition and Conversion Standards	Data Acquisition and Conversion Standards	This could be customer standards, if Oracle standards are not used.

RA.160	BI Apps=>Rec. Custom BI=>Rec.	Conduct Business Data Source Gap Analysis	Data Source Gap Matrix	This may be required when universal adapter files or standard extract files need to be created by the customer.
CV.027.1	BI Apps=>Rec. Custom BI=>Rec.	Perform Data Mapping	Data Mapping	This task most likely will require updating mapping for flex fields to satisfy new requirements.
RA.170.1	BI Apps=>Opt. Custom BI=>Rec.	Conduct Data Quality Assessment	High-Level Data Quality Assessment	This is usually a customer responsibility.
CV.035.1	BI Apps=>Opt. Custom BI=>Rec.	Define Manual Conversion Procedures (Initial Load)	Manual Conversion Procedures (Initial Load)	
CV.040.1	BI Apps=>Opt. Custom BI=>Rec.	Design Conversion Components (Initial Load)	Conversion Component Designs (Initial Load)	This may be the same as ongoing interfaces only with higher volumes.
CV.050.1	BI Apps=>Opt. Custom BI=>Rec.	Prepare Conversion Test Plans (Initial Load)	Conversion Test Plans (Initial Load)	
CV.055.1	BI Apps=>Opt. Custom BI=>Rec.	Implement Conversion Components (Initial Load)	Conversion Components (Initial Load)	
CV.060.1	BI Apps=>Opt. Custom BI=>Rec.	Perform Conversion Component Unit Test (Initial Load)	Unit-Tested Conversion Components (Initial Load)	
CV.062.1	BI Apps=>Opt. Custom BI=>Rec.	Perform Conversion Component Business Object Test (Initial Load)	Business Object-Tested Conversion Components (Initial Load)	
CV.063.1	BI Apps=>Opt. Custom BI=>Rec.	Perform Conversion Component Validation Test (Initial Load)	Validation-Tested Conversion Components (Initial Load)	

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B.ACT.PBPII Prepare Business Process Impact Inventory

BI Area of Focus	**Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)**
BI Applications	Optional
Custom BI	Optional

This activity is part of OUM's comprehensive set of Organizational Change Management materials that help to prepare the organization for the adoption of the new system. This work is frequently the responsibility of the customer and should always be scaled to suit the needs of the project.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
OCM.160	BI Apps=>Opt. Custom BI=>Opt.	Prepare Business Process Impact Inventory	Business Process Impact Inventory	

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B.ACT.DCMRCCE Deploy Change Management Roadmap and Communication Campaign - Elaboration

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Optional

This activity is part of OUM's comprehensive set of Organizational Change Management materials that help to prepare the organization for the adoption of the new system. This work is frequently the responsibility of the customer and should always be scaled to suit the needs of the project.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
OCM.150.2	BI Apps=>Opt. Custom BI=>Opt.	Conduct Change Management Roadmap and Communication Campaign	Change Management Roadmap and Communication Events	
OCM.155.1	BI Apps=>Opt. Custom BI=>Opt.	Monitor Change Management Roadmap and Communication Campaign Effectiveness	Change Management Roadmap and Communication Campaign Effectiveness Assessment	

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B.ACT.LA Lifecycle Architecture Milestone

For BI projects, the following recommended activities should be completed in order to achieve this milestone in the project:

Activity ID	Activity Name	BI Applications (including EPM)	Custom BI
B.ACT.DPS	Define Project Strategy	Recommended	Recommended

B.ACT.DI	Define Infrastructure	Recommended	Recommended
B.ACT.PEE	Prepare Environments - Elaboration	Recommended	Recommended
B.ACT.GBRE	Gather Business Requirements - Elaboration	Recommended	Recommended
B.ACT.DTP	Develop Test Plans	Recommended	Recommended
B.ACT.SSC	Specify Software Configuration	Recommended	Not Applicable
B.ACT.PVC	Prototype and Validate Configuration	Recommended	Not Applicable
B.ACT.PGA	Perform Gap Analysis	Recommended	Not Applicable
B.ACT.DUCM	Develop Use Case Model	Recommended	Recommended
B.ACT.CCPE	Create Conceptual Prototype - Elaboration	Recommended	Recommended
B.ACT.CS	Consolidate Specification	Recommended	Recommended
B.ACT.BSA	Baseline Software Architecture	Recommended	Recommended
B.ACT.DUCM	Develop Use Case Model	Recommended	Recommended
B.ACT.AE	Analyze - Elaboration	Recommended	Recommended
B.ACT.DE	Design - Elaboration	Recommended	Recommended
B.ACT.DCCTSE	Develop Custom Component Test Scripts - Elaboration	Optional	Recommended
B.ACT.DSTSE	Develop System Test Scenarios - Elaboration	Optional	Recommended
B.ACT.DVCSP	Develop and Validate Custom Software Prototypes	Recommended	Recommended
B.ACT.PCCTE	Perform Custom Component Testing - Elaboration	Optional	Recommended
B.ACT.PSTE	Perform System Test - Elaboration	Optional	Recommended
B.ACT.PPM	Plan Performance Management	Optional	Optional
B.ACT.PACDE	Prepare to Acquire and Convert Data - Elaboration	Recommended	Recommended
B.ACT.PBPII	Perform Business Process Impact Inventory	Optional	Optional
B.ACT.DCMRCCE	Deploy Change Management Roadmap and Communication Campaign - Elaboration	Optional	Optional

Note: When the optional activities listed above are applicable to your BI project, it is recommended that you complete them in order to achieve this milestone.

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CONSTRUCTION

C.ACT.FR Finalize Requirements

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

This activity is all about insuring that all critical documentation has been finalized as agreed upon in the contract. It is important to note that many times interim work

products become outdated as subsequent work products are developed to produce the ultimate work product, which is the software solution. This activity is where one reviews critical documentation, and updates it as necessary or required by original agreements in the contract.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
RD.042.3	BI Apps=>Rec. Custom BI=>Rec.	Develop Glossary	Glossary	Consider keeping a Project Notebook with the signed/approved final versions of the key deliverable work products.
RD.045.3	BI Apps=>Rec. Custom BI=>Rec.	Prioritize Requirements (MoSCoW)	MoSCoW List	
RD.130.2	BI Apps=>Rec. Custom BI=>Rec.	Develop Baseline Architecture Description	Architecture Description	
RA.023.3	BI Apps=>Rec. Custom BI=>Rec.	Develop Use Case Model	Use Case Model	
RA.024.2	BI Apps=>Rec. Custom BI=>Rec.	Develop Use Case Details	Use Case Specification	
RA.027.3	BI Apps=>Rec. Custom BI=>Rec.	Identify Candidate Business Rules	Candidate Business Rules	
RA.028.3	BI Apps=>Opt. Custom BI=>Rec.	Populate Business Rules Repository	Populated Business Rules Repository	Business rules may exist in the ETL or metadata calculations.
RA.180.2	BI Apps=>Rec. Custom BI=>Rec.	Review Use Case Model	Results Reviewed	Use meeting minutes.

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C.ACT.AC Analyze - Construction

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

This activity is generally performed when there is a need to gain further clarification of the business requirements represented in the use cases or there are multiple

iterations groups of use cases being developed for custom components. This typically applies to those requirements that can only be satisfied by custom-built components, which extend the functionality of the BI and EPM system and/or integrate the BI and EPM system with other BI and EPM or legacy systems. Finalizing requirements may lead to additional analysis and design work.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
AN.040.2	BI Apps=>Rec. Custom BI=>Rec.	Develop Analysis Architecture Description	Architecture Description	
AN.050.2	BI Apps=>Rec. Custom BI=>Rec.	Analyze Data	Analysis Model	This could impact ETL or meta data.
AN.060.2	BI Apps=>Rec. Custom BI=>Rec.	Analyze Behavior	Behavior Analysis	
AN.070.2	BI Apps=>Rec. Custom BI=>Rec.	Analyze Business Rules	Business Rules Analysis	
AN.090.2	BI Apps=>Rec. Custom BI=>Rec.	Analyze User Interface	User Interface Analysis	
AN.100.2	BI Apps=>Opt. Custom BI=>Rec.	Prepare Analysis Specification	Analysis Specification	
AN.110.2	BI Apps=>Rec. Custom BI=>Rec.	Review Analysis Model	Results Reviewed	Always document review sessions to capture decisions such as confirming items, and/or changed items.

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C.ACT.DC Design - Construction

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

This activity is generally performed when there is a need to gain further clarification of the business requirements represented in the use cases or there are multiple iterations groups of use cases being developed for custom components. Finalizing requirements may lead to additional analysis and design work.

	Task			

Task ID	Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
DS.040.2	BI Apps=>Rec. Custom BI=>Rec.	Develop Design Architecture Description	Architecture Description	
DS.080.2	BI Apps=>Rec. Custom BI=>Rec.	Design Software Components	Software Component Design	
DS.090.2	BI Apps=>Rec. Custom BI=>Rec.	Design Data	Component Data Design	
DS.100.2	BI Apps=>Rec. Custom BI=>Rec.	Design Behavior	Component Behavior Design	
DS.110.2	BI Apps=>Rec. Custom BI=>Rec.	Design Business Rules	Business Rules Design	
DS.130.2	BI Apps=>Rec. Custom BI=>Rec.	Design User Interface	User Interface Design	
DS.140.2	BI Apps=>Rec. Custom BI=>Rec.	Prepare Design Specification	Design Specification	
DS.150.2	BI Apps=>Rec. Custom BI=>Rec.	Develop Database Design	Logical Database Design	Document changes to the model in most cases.
DS.160.2	BI Apps=>Rec. Custom BI=>Rec.	Review Design Model	Results Reviewed	Use meeting minutes.

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C.ACT.PTP Perform Test Planning

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

The Perform Test Planning activity allows for updating and refinement of the Integration and System Test Plans.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
TE.015.2	BI Apps=>Rec. Custom BI=>Rec.	Develop Integration Test Plan	Integration Test Plan	The scope of testing may differ for an EPM and a BI Apps project.
TE.050.2	BI Apps=>Rec. Custom BI=>Rec.	Develop System Test Plan	System Test Plan	

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C.ACT.PEC Prepare Environments - Construction

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

This activity addresses the installation of software environments for testing and development. In this activity the “out of the box” software is installed. Set up of environments may occur earlier in the development cycle

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
IM.007.2	BI Apps=>Rec. Custom BI=>Rec.	Prepare Development Environment	Development Environment	
TE.018.2	BI Apps=>Rec. Custom BI=>Rec.	Prepare Static Test Data	Static Test Data	This is usually integrated with the customer source development system(s).
TE.038.2	BI Apps=>Opt. Custom BI=>Rec.	Prepare Integration Test Environment	Integration Test Environment	There is usually only one test environment, i.e., the same environment is used for all testing

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C.ACT.DCCTSC Develop Custom Component Test Scripts - Construction

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Recommended

This activity is performed for all custom-built components that extend the functionality of EPM or BI Apps system. Actual work related to this particular activity and tasks are tied directly to the building of the components that occurs in IM.050.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
TE.020.2	BI Apps=>Opt. Custom BI=>Rec.	Develop Unit Test Scripts	Unit Test Checklists	
TE.035.2	BI Apps=>Opt. Custom BI=>Rec.	Create Integration Test Scenarios	Integration Test Scenarios	Develop integration test scenarios where complex logic between related modules exist. These test scenarios most likely are appropriate to include in the system test scenarios.

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C.ACT.DSTSC Develop System Test Scenarios – Construction

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

This activity is performed for all custom-built components that extend the functionality of the EPM and BI Apps system.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
TE.025.3	BI Apps=>Rec. Custom BI=>Rec.	Create System Test Scenarios	System Test Scenarios	

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C.ACT.DSITS Develop Systems Integration Test Scenarios

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Optional

During this activity, you develop the Systems Integration Test Scenarios that will be used when performing the Systems Integration Test. It is recommended if the BI Apps or Custom BI system integrates with other systems.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
TE.090	BI Apps=>Opt. Custom BI=>Opt.	Develop Systems Integration Test Scenarios	Systems Integration Test Scenarios	

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C.ACT.ICC Implement Custom Components

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

This activity allows for implementing (i.e., building) the custom components and custom services needed to extend the capabilities of the application system. It also provides for updating/finalizing the application setup parameters in preparation for Perform System Test (TE.070).

The Perform Component Review task is often completed for EPM - BI Apps projects. In this activity you make the entire product parts (out of box and custom) communicate with each other. The technical staff conducts an end to end test.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
IM.040.2	BI Apps=>Rec. Custom BI=>Rec.	Implement Database	Physical Database Design	
IM.050	BI Apps=>Rec.	Implement Components	Implemented Components	

	Custom BI=>Rec.			
IM.055.2	BI Apps=>Opt. Custom BI=>Rec.	Perform Business Rules Implementation (Rules Engine)	Implemented Business Rules (Rules Engine)	This is usually part of ETL or metadata.
IM.060.2	BI Apps=>Rec. Custom BI=>Rec.	Perform Component Review	Results Reviewed	
IM.070	BI Apps=>Rec. Custom BI=>Rec.	Assemble Components	Assembled Components	
IM.080	BI Apps=>Opt. Custom BI=>Rec.	Integrate Services	Integrated Services	
IM.090	BI Apps=>Rec. Custom BI=>N/A	Create Installation Routines	Installation Instructions	This could be a stand-alone document, or the appendix of the Installation Plan.
TE.065	BI Apps=>Rec. Custom BI=>Rec.	Perform Installation Test	Tested Installation Routines	

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C.ACT.PCCTC Perform Custom Component Testing - Construction

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

This activity is performed for all custom-built components that extend the functionality of EPM or BI Apps system. Actual work related to this particular activity and tasks are tied directly to the building of the components that occurs in IM.050. In BI, the unit test actually is an integration test with the source system.

The OUM BI view testing approach is based upon four types of testing:

Unit Testing - Testing that occurs when one develops individual components of the solution (covered in this activity).

Integration Testing - Testing that occurs between some components, also known as string testing. Simply makes sure associated components fulfilling a common functionality are working correctly together (not to be confused with System testing).

System Testing - End-to-end testing of the complete solution, which includes running of any batch cycles. Last chance to verify/validate system prior to users testing.

Acceptance Testing - Also known as user acceptance testing. This is where the system is tested to verify it meets the originally agreed upon acceptance criteria.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
TE.030.2	BI Apps=>Rec. Custom BI=>Rec.	Perform Unit Test	Unit-Tested Components	All testing results contribute to the ultimate goal of a fully functioning, tested solution, which is TE.070. However, the contract may require a stand-alone document to be produced for each stage of testing.
TE.040.2	BI Apps=>Rec. Custom BI=>Rec.	Perform Integration Test	Integration Test Results	All testing results contribute to the ultimate goal of a fully functioning, tested solution, which is TE.070. However, the contract may require a stand-alone document to be produced for each stage of testing.

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C.ACT.PST Prepare for System Test

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

n this activity, you prepare for the System Test. If neccessary, apply any updates to the BI Apps configuration based on the Application Setup Documents (MC.050) and prepare the System Test Environment.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
MC.050.2	BI Apps=>Rec. Custom BI=>N/A	Define Application Setups	Application Setup Documents	This may only require standard technical installation guides.
TE.060.2	BI Apps=>Rec. Custom BI=>Rec.	Prepare System Test Environment	System Test Environment	There is usually only one test environment, i.e., the same environment is used for all testing

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C.ACT.PSTC Perform System Test – Construction

	Activity

BI Area of Focus	Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

This activity is performed for all custom-built components that extend the functionality of the EPM and BI Apps system and/or integrate the EPM and/or BI Apps system with other EPM and BI Apps or legacy systems. In BI, the unit test actually is an integration test with the source system.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
TE.070.2	BI Apps=>Rec. Custom BI=>Rec.	Perform System Test	System Test Results	
RA.055.2	BI Apps=>Opt. Custom BI=>N/A	Document Business Procedures	Business Procedure Documentation	
TE.019.1	BI Apps=>Opt. Custom BI=>Opt.	Collect, Assess, and Refine KPI Measurements	Key Business Strategies and Objectives	

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C.ACT.PSIT Perform Systems Integration Test

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Optional

Systems Integration testing is necessary to consider if the solution is feeding information to another system that is not part of the solution being built. It is important to note that supporting systems that are utilized by the system do not necessarily constitute the need for a Systems Integration Test. An example of this would be when a BI solution is obtaining processing currency rate data feeds from third-party vendor during the transformation process.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
TE.085	BI Apps=>Opt. Custom BI=>Opt.	Prepare Systems Integration Test Environment	System Integration Test Environment	There is usually only one test environment, i.e., the same environment is used for all testing

TE.100	BI Apps=>Opt. Custom BI=>Opt.	Perform Systems Integration Test	Systems Integration Test Results	

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C.ACT.PPT Prepare for Performance Testing

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Optional

The Prepare for Performance Testing activity consists of building the test scripts and programs, constructing the environment and database and conducting a dress rehearsal.

Note: There will always be some performance requirements, but this activity would only be used if there was a formal performance testing requirement agreed upon in the contract.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
PT.070	BI Apps=>Opt. Custom BI=>Opt.	Build Performance Test Scripts and Programs	Performance Test Scripts and Programs	
PT.080	BI Apps=>Opt. Custom BI=>Opt.	Construct Performance Test Environment and Database	Performance Test Environment	
PT.090	BI Apps=>Opt. Custom BI=>Opt.	Conduct Performance Test Dress Rehearsal	Validated Performance Test and Environment	

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C.ACT.PT Prepare for Transition

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI	Recommended

Applications	
Custom BI	Recommended

This activity consists of defining the System Management Procedures, Operational Test Plan, and the Final Platform and Network Architecture and the Capacity Plan.

It is recommended that the Define Final Platform and Network Architecture task is completed for this activity for BI and EPM implementation projects.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
TA.100	BI Apps=>Rec. Custom BI=>Rec.	Define System Management Procedures	System Management Guide	
TA.110	BI Apps=>Opt. Custom BI=>Rec.	Define Operational Testing Plan	Operational Testing Plan	
TA.150	BI Apps=>Rec. Custom BI=>Rec.	Define Final Platform and Network Architecture	Final Platform and Network Architecture	
TA.160	BI Apps=>Rec. Custom BI=>Rec.	Define System Capacity Plan	System Capacity Plan	Contract may require a stand-alone document.

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C.ACT.PCO Prepare for Cutover

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

This activity consists of defining the Cutover Strategy, developing the Installation Plan, and designing the Production Support Infrastructure.

All tasks within this activity are recommended for BI and EPM implementation projects.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
TS.020.2	BI Apps=>Rec.	Define Cutover Strategy	Cutover Strategy	

	Custom BI=>Rec.			
TS.030	BI Apps=>Rec. Custom BI=>Rec.	Develop Installation Plan	Installation Plan	
TS.040	BI Apps=>Rec. Custom BI=>Rec.	Design Production Support Infrastructure	Production Support Infrastructure Design	

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C.ACT.TI Test Infrastructure

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

This activity consists of conducting operational, backup and recovery and disaster recovery testing.

All tasks within this activity are recommended for BI and EPM implementation projects.

Note: The tasks in this activity are normally owned by the customer, unless otherwise specified.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
TA.120	BI Apps=>Rec. Custom BI=>Rec.	Conduct Operational Testing	Operational Test Results	
TA.130	BI Apps=>Opt. Custom BI=>Rec.	Conduct Backup and Recovery Test	Backup and Recovery Test Results	
TA.140	BI Apps=>Opt. Custom BI=>Rec.	Conduct Disaster Recovery Test	Disaster Recovery Test Results	This task is rare to be in scope. It would normally be a stand-alone project.

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C.ACT.PACDC Prepare to Acquire and Convert Data – Construction

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

Although this activity is almost always required in BI solutions, it will generally be performed as one of the component implementation tasks.

All tasks within this activity are recommended for BI and EPM implementation projects if data is to be migrated from a legacy system.

This task evolves during the development of a project. It begins with the identification of the source systems and evolves into the actual data source of the attributes needed in the target system. In many cases, some type of transformation is needed to conform to the target system's architecture. Some data elements needed in the target system may not be available. These elements may be entered manually through the application or in some cases, a new function is required to maintain the "missing" source data elements. The final source to target mappings with the transformations should be documented for BI and EPM implementation projects.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
CV.027.2	BI Apps=>Rec. Custom BI=>Rec.	Perform Data Mapping	Data Mapping	
RA.170.2	BI Apps=>Opt. Custom BI=>Rec.	Conduct Data Quality Assessment	High-Level Data Quality Assessment	
CV.035.2	BI Apps=>Rec. Custom BI=>Rec.	Define Manual Conversion Procedures (Initial Load)	Manual Conversion Procedures (Initial Load)	
CV.040.2	BI Apps=>Rec. Custom BI=>Rec.	Design Conversion Components (Initial Load)	Conversion Component Designs (Initial Load)	
CV.050.2	BI Apps=>Opt. Custom BI=>Rec.	Prepare Conversion Test Plans (Initial Load)	Conversion Test Plans (Initial Load)	

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C.ACT.ACDC Acquire and Convert Data – Construction

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended

Custom BI	Recommended

Although this activity is almost always required in BI solutions, it will generally be performed as one of the component implementation tasks.

All tasks within this activity are recommended for BI and EPM implementation projects if data is to be migrated from a legacy system.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
CV.030.2	BI Apps=>Opt. Custom BI=>Rec.	Prepare Conversion Environment (Initial Load)	Conversion Environment (Initial Load)	
CV.055.2	BI Apps=>Rec. Custom BI=>Rec.	Implement Conversion Components (Initial Load)	Conversion Components (Initial Load)	
CV.060.2	BI Apps=>Opt. Custom BI=>Rec.	Perform Conversion Component Unit Test (Initial Load)	Unit-Tested Conversion Components (Initial Load)	
CV.062.2	BI Apps=>Opt. Custom BI=>Rec.	Perform Conversion Component Business Object Test (Initial Load)	Business Object-Tested Conversion Components (Initial Load)	
CV.063.2	BI Apps=>Rec. Custom BI=>Rec.	Perform Conversion Component Validation Test (Initial Load)	Validation-Tested Conversion Components (Initial Load)	
CV.064.1	BI Apps=>Rec. Custom BI=>Rec.	Install Conversion Components (Initial Load)	Installed Conversion Components (Initial Load)	
CV.065.1	BI Apps=>Rec. Custom BI=>Rec.	Convert and Verify Data (Initial Load)	Converted and Verified Data (Initial Load)	
CV.068.1	BI Apps=>Opt. Custom BI=>Rec.	Clean Data	Clean Legacy Data	

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C.ACT.PD Produce Documentation

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended

Custom BI	Recommended

This activity consists of publishing the appropriate documentation and producing the Online Help. During the Transition Phase, you will have an opportunity to review all implemented system changes and system defects for any impact on the Documentation work products and finalize these work products.

Note: Documentation needs to match contract and PMP scope requirements

Documentation is often requested for EPM and BI Apps WCM projects – If documentation has been requested all tasks in this activity should be completed.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
DO.060	BI Apps=>Opt. Custom BI=>Rec.	Publish User Reference Manual	User Reference Manual	This is typically addressed by customer.
DO.070	BI Apps=>Opt. Custom BI=>Rec.	Publish User Guide	User Guide	This is typically addressed by customer.
DO.080	BI Apps=>Rec. Custom BI=>Rec.	Publish Technical Reference Material	Technical Reference Material	
DO.100	BI Apps=>Opt. Custom BI=>Rec.	Produce Online Help	Online Help	This is part of the general design.

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C.ACT.DCMRCCC Deploy Change Management Roadmap and Communication Campaign - Construction

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Optional

This activity is part of OUM's comprehensive set of Organizational Change Management materials that help to prepare the organization for the adoption of the new system. This work is frequently the responsibility of the customer and should always be scaled to suit the needs of the project.

Task ID	Task Consideration (Link to BI Supp Task	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes

	Guidance)			
OCM.150.3	BI Apps=>Opt. Custom BI=>Opt.	Conduct Change Management Roadmap and Communication Campaign	Change Management Roadmap and Communication Campaign Effectiveness Assessment	
OCM.155.2	BI Apps=>Opt. Custom BI=>Opt.	Monitor Change Management Roadmap and Communication Campaign Effectiveness	Change Management Roadmap and Communication Campaign Effectiveness Assessment	

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C.ACT.CJIA Conduct Job Impact Analysis

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Optional

This activity is part of OUM's comprehensive set of Organizational Change Management materials that help to prepare the organization for the adoption of the new system. This work is frequently the responsibility of the customer and should always be scaled to suit the needs of the project.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
OCM.170	BI Apps=>Opt. Custom BI=>Opt.	Collect and Analyze Job Change Data	Job Impact Analysis	
OCM.180	BI Apps=>Opt. Custom BI=>Opt.	Determine Impact of Job Changes	Job Impact Diagnosis	
OCM.190	BI Apps=>Opt. Custom BI=>Opt.	Prepare HR Transition Plan	HR Transition Plan	

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C.ACT.CMAWC Conduct Managers Alignment Workshop - Construction

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)

BI Applications	Optional
Custom BI	Optional

This activity is part of OUM's comprehensive set of Organizational Change Management materials that help to prepare the organization for the adoption of the new system. This work is frequently the responsibility of the customer and should always be scaled to suit the needs of the project.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
OCM.200	BI Apps=>Opt. Custom BI=>Opt.	Design Managers Unit and Department Impact Workshop	Managers Unit and Department Impact Workshop Materials	
OCM.210	BI Apps=>Opt. Custom BI=>Opt.	Conduct Managers Unit and Department Impact Workshop	Enabled Managers	

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C.ACT.DEUT Design End-User Training

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Recommended

The Design End-User Training activity consists of executing the Training tasks to design the training for the users for the new BI and EPM system.

In an BI and EPM system implementation, user learning events are typically conducted in the Transition Phase.

Note: This is normally the customer's responsibility.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
TR.060	BI Apps=>Opt. Custom BI=>Rec.	Conduct User Learning Needs Analysis	User Learning Needs Analysis	
TR.070	BI Apps=>Opt. Custom BI=>Rec.	Develop User Learning Plan	User Learning Plan	

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C.ACT.BEUT Build End-User Training

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Recommended

The Build End-User Training activity consists of executing the Training tasks to build the training for the users for the new BI and EPM system.

In an BI and EPM system implementation, user learning events are typically conducted in the Transition Phase.

Note: This is normally the customer's responsibility.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
TR.080	BI Apps=>Opt. Custom BI=>Rec.	Develop User Learningware	User Learningware	
TR.090	BI Apps=>Opt. Custom BI=>Rec.	Prepare User Learning Environment	Configured User Learning Environment	

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C.ACT.TEUC Train End Users - Construction

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Recommended

The Train End Users activity consists of executing the Training tasks for training the users for the new BI and EPM system.

In an BI and EPM system implementation, user learning events are typically conducted in the Transition Phase.

Note: This is normally the customer's responsibility.

	Task			

Task ID	Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
TR.100.1	BI Apps=>Opt. Custom BI=>Rec.	Conduct User Learning Events	User Learning Events Administration	

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C.ACT.IOC Initial Operational Capability Milestone

For BI projects, the following recommended activities should be completed in order to achieve this milestone in the project:

Activity ID	Activity Name	BI Applications (including EPM)	Custom BI
C.ACT.FR	Finalize Requirements	Recommended	Recommended
C.ACT.AC	Analyze - Construction	Recommended	Recommended
C.ACT.DC	Design - Construction	Recommended	Recommended
C.ACT.PTP	Perform Test Planning	Recommended	Recommended
C.ACT.PEC	Prepare Environments - Construction	Recommended	Recommended
C.ACT.DCCTSC	Develop Custom Component Test Scripts - Construction	Optional	Recommended
C.ACT.DSTSC	Develop System Test Scenarios - Construction	Recommended	Recommended
C.ACT.DSITS	Develop Systems Integration Test Scenarios	Optional	Optional
C.ACT.ICC	Implement Custom Components	Recommended	Recommended
C.ACT.PCCTC	Perform Custom Component Testing - Construction	Recommended	Recommended
C.ACT.PST	Prepare for System Test	Recommended	Recommended
C.ACT.PSTC	Perform System Test - Construction	Recommended	Recommended
C.ACT.PSIT	Perform Systems Integration Test	Optional	Optional
C.ACT.PPT	Prepare for Performance Testing	Optional	Optional
C.ACT.PT	Prepare for Transition	Recommended	Recommended
C.ACT.PCO	Prepare for Cutover	Recommended	Recommended
C.ACT.TI	Test Infrastructure	Recommended	Recommended
C.ACT.PACDC	Prepare to Acquire and Convert Data - Construction	Recommended	Recommended
C.ACT.ACDC	Acquire and Convert Data - Construction	Recommended	Recommended
C.ACT.PD	Produced Documentation	Recommended	Recommended
C.ACT.DCMRCCC	Deploy Change Management Roadmap and Communication Campaign - Construction	Optional	Optional
C.ACT.CJIA	Conduct Job Impact Analysis	Optional	Optional
C.ACT.CMAWC	Conduct Managers Alignment Workshop - Construction	Optional	Optional

C.ACT.DEUT	Design End-User Training	Optional	Recommended
C.ACT.BEUT	Build End-User Training	Optional	Recommended
C.ACT.TEUC	Train End Users - Construction	Optional	Recommended

Note: When the optional activities listed above are applicable to your BI project, it is recommended that you complete them in order to achieve this milestone.

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TRANSITION

D.ACT.SUAT Support User Acceptance Test

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

This activity consists of preparing users for testing, preparing the Acceptance Test Environment and supporting the users in conducting the acceptance test.

This environment is commonly known as the User Acceptance Testing (UAT) environment and is used for many BI and EPM implementation projects.

Although the tasks in this activity are optional, it is strongly recommended that user testing is conducted.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
TE.105	BI Apps=>Opt. Custom BI=>Rec.	Prepare Users for Testing	Prepared Users for Testing	
TE.110	BI Apps=>Opt. Custom BI=>Rec.	Prepare Acceptance Test Environment	Acceptance Test Environment	
TE.120	BI Apps=>Rec. Custom BI=>Rec.	Support Acceptance Test	Acceptance Test Results	

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D.ACT.CPTST Conduct Performance Test

	Activity

BI Area of Focus	Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Optional

The Conduct Performance Test activity consists of executing the performance test and creating the Performance Test Results. Some performance testing is always done, but this is a detailed performance test activity that must be accounted for in the estimate and the contract, if performed.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
PT.100	BI Apps=>Opt. Custom BI=>Opt.	Execute Performance Test	Performance Test Results	
PT.110	BI Apps=>Opt. Custom BI=>Opt.	Create Performance Test Report	Performance Test Report	

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D.ACT.PPE Prepare Production Environment

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

The Go Production activity consists of preparing the Production Environment, going production, and connecting to the source systems.

All tasks within this activity are recommended for BI and EPM implementation projects.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
TS.050	BI Apps=>Rec. Custom BI=>Rec.	Prepare Production Environment	Production Environment	

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D.ACT.CDT Convert Data – Transition

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Optional

The objective for this activity is to convert and cleanse data / content in preparation for the transition of the solution into the Production Environment. this is generally needed, if there is an initial load process.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
CV.064.2	BI Apps=>Opt. Custom BI=>Rec.	Install Conversion Components (Initial Load)	Installed Conversion Components (Initial Load)	This should include software moves to production.
CV.065.2	BI Apps=>Opt. Custom BI=>Rec.	Convert and Verify Data (Initial Load)	Converted and Verified Data (Initial Load)	
CV.068.2	BI Apps=>Opt. Custom BI=>Rec.	Clean Data	Clean Legacy Data	Data quality is a customer responsibility, unless specified in contract.

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D.ACT.DCMRCCT Deploy Change Management Roadmap and Communication Campaign - Transition

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Optional

This activity is part of OUM's comprehensive set of Organizational Change Management materials that help to prepare the organization for the adoption of the new system. This work is frequently the responsibility of the customer and should always be scaled to suit the needs of the project.

Task ID	Task Consideration (Link to BI Supp Task	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes

	Guidance)			
OCM.150.4	BI Apps=>Opt. Custom BI=>Opt.	Conduct Change Management Roadmap and Communication Campaign	Change Management Roadmap and Communication Campaign Effectiveness Assessment	
OCM.155.3	BI Apps=>Opt. Custom BI=>Opt.	Monitor Change Management Roadmap and Communication Campaign Effectiveness	Change Management Roadmap and Communication Campaign Effectiveness Assessment	

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D.ACT.CITA Conduct IT Alignment

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Optional

This activity is part of OUM's comprehensive set of Organizational Change Management materials that help to prepare the organization for the adoption of the new system. This work is frequently the responsibility of the customer and should always be scaled to suit the needs of the project.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
OCM.220	BI Apps=>Opt. Custom BI=>Opt.	Prepare Assessment of Impact on IT Groups	IT Alignment Diagnosis Grid	
OCM.230	BI Apps=>Opt. Custom BI=>Opt.	Prepare IT Transition Plan	IT Transition Plan	

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D.ACT.TEUT Train End Users - Transition

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Recommended

The Train End Users activity consists of executing the Training tasks for training the users for the new BI and EPM system.

In a BI and EPM implementation, user learning events are typically conducted in the Transition Phase.

Note: This is normally a customer responsibility.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
TR.100.2	BI Apps=>Opt. Custom BI=>Rec.	Conduct User Learning Events	User Learning Events Administration	

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D.ACT.FD Finalize Documentation

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

The Finalize Documentation activity provides a last opportunity to make any revisions and finalize documentation and Online Help.

Note: Only finalize documentation to levels that were agreed upon in the contract. There will most likely be a great deal of interim documentation that will have become outdated, as its results are used in later activities/tasks, and keeping past items updated with changes resulting from later work can require a great deal of time and effort...Only do that which is necessary to build the solution, or to meet the contractual obligations.

All tasks in this activity are recommended for BI and EPM system implementation projects.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
DO.110	BI Apps=>Rec. Custom BI=>Rec.	Finalize Documentation	Final Documentation Work Products	It is important to note that documentation is finalized only to the state to which it is specified in the contract.

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D.ACT.GP Go Production

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

The Go Production activity consists of preparing the Production Environment, going production, and connecting to the source systems.

All tasks within this activity are recommended for BI and EPM implementation projects.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
TS.052	BI Apps=>Rec. Custom BI=>Rec.	Implement Production Support Infrastructure	Production Support Infrastructure	
TS.058	BI Apps=>Rec. Custom BI=>Rec.	Verify Production Readiness	Production-Ready System	
TS.060	BI Apps=>Rec. Custom BI=>Rec.	Go Production	System in Production	
TS.070	BI Apps=>Opt. Custom BI=>Rec.	Shut Down Legacy System	Legacy System Shut Down	This task is nearly always owned by customer.

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D.ACT.SP System in Production Milestone

For BI projects, the following recommended activities should be completed in order to achieve this milestone in the project:

Activity ID	Activity Name	BI Applications (including EPM)	Custom BI
D.ACT.SUAT	Support User Acceptance Test	Recommended	Recommended
D.ACT.CPTST	Perform Performance Test	Optional	Optional
D.ACT.PPE	Prepare Production Environment	Recommended	Recommended
D.ACT.CDT	Convert Data - Transition	Optional	Recommended
D.ACT.DCMRCCT	Deploy Change Management Roadmap and Communication Campaign - Transition	Optional	Optional
D.ACT.CITA	Conduct IT Alignment	Optional	Optional
D.ACT.TEUT	Train End Users - Transition	Optional	Recommended

D.ACT.FD	Finalize Documentation	Recommended	Recommended
D.ACT.GP	Go Production	Recommended	Recommended

Note: When the optional activities listed above are applicable to your BI project, it is recommended that you complete them in order to achieve this milestone.

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PRODUCTION

E.ACT.MPSP Manage Production System Performance

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Optional

The Manage Production System Performance activity provides for managing the Production Environment. The objective for this activity is to monitor and adjust the BI and EPM system based on production performance.

Note: This activity is only performed if it is part of the contract, and time was originally estimated for it.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
PT.120	BI Apps=>Opt. Custom BI=>Opt.	Conduct Production Performance Management	Managed Production Environment	

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E.ACT.EPS Evaluate Production System

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Recommended

The Evaluate Production System activity consists of auditing the system and analyzing any problems. The objective for this activity is to audit the EPM - BI Apps system after the move to production and to analyze any problems should they occur.

Note: This activity is only performed if it is part of the contract, and time was originally estimated for it.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
TE.019.2	BI Apps=>Opt. Custom BI=>Rec.	Collect, Assess and Refine KPI Measurements	Key Business Strategies and Objectives	
PS.010	BI Apps=>Opt. Custom BI=>Rec.	Audit System	System Evaluation	

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E.ACT.RPP Resolve Production Problems

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Recommended

The Resolve Production Problems activity consists of resolving production problems. The ongoing system maintenance begins during this activity. Maintenance is typically not triggered by technical issues that the support or help desk address; this type of maintenance focuses on necessary and preventative actions. Ongoing system maintenance should address the following:

- timing of maintenance procedures reporting unusual occurrences interpreting system indicators logging completion of maintenance work
- adjusting the software and hardware for anticipated business changes

Verify that all normal, ongoing activities keep the system protected, purged, and operating as efficiently as possible. Make sure a log is available for recording all change events. Standard checklists can be incorporated for normal, standing tasks that must be performed.

Note: This activity is only performed if it is part of the contract, and time was originally estimated for it.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
PS.050	BI Apps=>Opt. Custom BI=>Rec.	Analyze Problems	Fault Corrections	
PS.060	BI Apps=>Opt. Custom	Monitor and Respond to System Problems	Problem Log	

	BI=>Rec.			

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E.ACT.DCMRCCP Deploy Change Management Roadmap and Communication Campaign - Production

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Optional

This activity is part of OUM's comprehensive set of Organizational Change Management materials that help to prepare the organization for the adoption of the new system. This work is frequently the responsibility of the customer and should always be scaled to suit the needs of the project.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
OCM.150.5	BI Apps=>Opt. Custom BI=>Opt.	Conduct Change Management Roadmap and Communication Campaign	Change Management. Roadmap and Communication Events	
OCM.155.4	BI Apps=>Opt. Custom BI=>Opt.	Monitor Change Management Roadmap and Communication Campaign Effectiveness	Change Management Roadmap and Communication Campaign Effectiveness Assessment	

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E.ACT.PF Plan for Future

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Optional

The Plan for Future activity consists of determining the Future Functional Enhancements and prioritizing them in the Future MoSCoW List as well as measuring the effectiveness of the organizational change effort and making post-Go Live recommendations. The purpose of this activity is to identify follow up work that may have been outlined in the initial proposal. One should review the MoSCoW List for those requirements that were classified as "should have" or "could have" that did not make it into this development release.

Although the tasks in this activity are optional, it is strongly recommended that some form of enhancement planning takes place.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
PS.135	BI Apps=>Opt. Custom BI=>Opt.	Determine Future Functional Enhancements	Future Functional Enhancements	
PS.140	BI Apps=>Opt. Custom BI=>Opt.	Plan Enhancements	Future MoSCoW List	
OCM.250	BI Apps=>Opt. Custom BI=>Opt.	Measure Organizational Change Effectiveness	Organizational Effectiveness Assessment	

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E.ACT.DITTP Deploy IT Transition Plan

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Optional
Custom BI	Optional

This activity is part of OUM's comprehensive set of Organizational Change Management materials that help to prepare the organization for the adoption of the new system. This work is frequently the responsibility of the customer and should always be scaled to suit the needs of the project.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
OCM.260	BI Apps=>Opt. Custom BI=>Opt.	Implement IT Transition Plan	Aligned IT Organization	

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E.ACT.SO Sign-Off Milestone

For BI Projects, the following recommended activities should be completed in order to achieve this milestone in the project:

Activity ID	Activity Name	BI Applications (including EPM)	Custom BI
E.ACT.MPSP	Manage Production System Performance	Optional	Optional
E.ACT.EPS	Evaluate Production System	Optional	Recommended
E.ACT.RPP	Resolve Production Problems	Optional	Recommended
E.ACT.DCMRCCP	Deploy Change Management Roadmap and Communication Campaign - Production	Optional	Optional
E.ACT.PF	Plan For Future	Optional	Optional
E.ACT.DITTP	Implement IT Transition Plan	Optional	Optional

Note: When the optional activities listed above are applicable to your BI project, it is recommended that you complete them in order to achieve this milestone.

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PROJECT CLOSURE

PC.ACT.GA Gain Acceptance

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

The Gain Acceptance activity includes gaining formal acceptance for the project from the customer for the BI and EPM implementation project.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
SM.050	BI Apps=>Rec. Custom BI=>Rec.	Gain Acceptance	Acceptance Certificate / Project Acceptance Report	Include this in the close-out report to customer.

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PC.ACT.CPC Close Processes and Contract

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended

Custom BI	Recommended

This activity closes out the project processes for each of the project management processes. In addition, the contract is reviewed to ensure that all obligations have been met.

It is recommended that all tasks for this activity are completed for BI and EPM implementation projects.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
SM.060	BI Apps=>Rec. Custom BI=>Rec.	Close Scope Management	Scope Management Lessons Learned	There should be a summary for Oracle management as well as in the customer close-out document.
SM.070	BI Apps=>Opt. Custom BI=>Opt.	Identify Future System Enhancements	Future Systems Enhancements	
FM.050	BI Apps=>Rec. Custom BI=>Rec.	Close Project Finals	Financial Management Lessons Learned	Project Close-Out
WM.060	BI Apps=>Rec. Custom BI=>Rec.	Close Work Management	Work Management Lessons Learned	Project Close-Out
RKM.060	BI Apps=>Opt. Custom BI=>Opt.	Conduct Post-Production Risk Assessment	Post-Production Risk Assessment / Risk Assessment Lessons Learned	Project Close-Out
IPM.050	BI Apps=>Opt. Custom BI=>Opt.	Produce Final Issue and Problem Report and Close Log(s)	Issue and Problem Management Lessons Learned	Project Close-Out
CMM.060	BI Apps=>Rec. Custom BI=>Rec.	Submit Final Reports		Project Close-Out - Project Close-Out Report
STM.070	BI Apps=>Rec. Custom BI=>Rec.	Release Staff	Released Staff	Project Close-Out
QM.060	BI Apps=>Opt. Custom BI=>Opt.	Conduct Post Production Quality Review	Customer Satisfaction Report / Project Health Checklist	Project Close-Out
CM.080	BI Apps=>Rec. Custom BI=>Rec.	Close Configuration Management	Configuration Management Lessons Learned	Project Close-Out
IFM.050	BI Apps=>Rec. Custom BI=>Rec.	Close Infrastructure	Closed Infrastructure	Project Close-Out
PCM.040	BI Apps=>Rec.	Close Contract	Supplier Assessment Record	Project Close-Out

	Custom BI=>Rec.			
OCHM.040	BI Apps=>Rec. Custom BI=>Rec.	Establish Follow-Up Process	Follow-Up Process	Project Close-Out

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PC.ACT.DLLAP Document Lessons Learned and Archive Project

BI Area of Focus	Activity Recommended / Optional (An activity is recommended if at least one task in the activity is recommended.)
BI Applications	Recommended
Custom BI	Recommended

The Document Lessons Learned and Archive Project activity compiles the lessons learned through project execution and control and it archives the project.

The tasks in this activity are recommended so that all information is captured for both Oracle and the customer.

Task ID	Task Consideration (Link to BI Supp Task Guidance)	Task Name (Link to OUM Base Task Overview)	Work Product Name	Task Notes
CMM.040	BI Apps=>Rec. Custom BI=>Rec.	Document Lessons Learned		Project Close-Out
CMM.050	BI Apps=>Rec. Custom BI=>Rec.	Turn Over Project Documentation (Project Archives)		Project Close-Out

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CUSTOM BUSINESS INTELLIGENCE (BI) SUPPLEMENTAL TASK GUIDELINES

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This document contains OUM supplemental task guidance for BI implementations, specifically custom BI implementations (not BI applications).

- The first column of the following table provides a complete list of all tasks in this document for which "supplemental" guidelines are provided as well as a quick link to the "supplemental" task guideline in this document.
- The second column provides a link to the OUM Task Overview.

The task guidelines in this document should be used to supplement the guidelines that appear in the OUM Task Overview.

BI - OUM Supplemental Task Guidelines (this document) collapse table | expand table	OUM Task Overview
Requirements Analysis	
RA.015 Develop Business Use Case Model	RA.015 Develop Business Use Case Model
RA.024 Develop Use Case Details	RA.024 Develop Use Case Details
RA.030 Validate Conceptual Prototype	RA.030 Validate Conceptual Prototype
RA.035 Develop High-Level Software Architecture Description	RA.035 Develop High-Level Software Architecture Description
Analysis	
AN.050 Analyze Data	AN.050 Analyze Data
AN.060 Analyze Behavior	AN.060 Analyze Behavior
AN.110 Review Analysis Model	AN.110 Review Analysis Model
Implementation	
IM.005 Develop Conceptual Prototype	IM.005 Develop Conceptual Prototype
IM.040 Implement Database	IM.040 Implement Database
IM.050 Implement Components	IM.050 Implement Components

IM.070 Assemble Components	IM.070 Assemble Components
Testing	
TE.005 Determine Testing Requirements	TE.005 Determine Testing Requirements
TE.010 Develop Testing Strategy	TE.010 Develop Testing Strategy
TE.015 Develop Integration Test Plan	TE.015 Develop Integration Test Plan
TE.020 Develop Unit Test Scripts	TE.020 Develop Unit Test Scripts
TE.050 Develop System Test Plan	TE.050 Develop System Test Plan
TE.080 Develop Systems Integration Test Plan	TE.080 Develop Systems Integration Test Plan
TE.110 Prepare Acceptance Test Environment	TE.110 Prepare Acceptance Test Environment
Performance Management	
PT.030 Define Performance Testing Strategy	PT.030 Define Performance Testing Strategy
PT.120 Conduct Production Performance Management	PT.120 Conduct Production Performance Management
Data Acquisition and Conversion	
CV.010 Define Data Acquisition and Conversion Requirements	CV.010 Define Data Acquisition and Conversion Requirements
CV.020 Define Data Acquisition, Conversion and Data Quality Strategy	CV.020 Define Data Acquisition, Conversion and Data Quality Strategy
CV.027 Perform Data Mapping	CV.027 Perform Data Mapping
CV.040 Design Conversion Components (Initial Load)	CV.040 Design Conversion Components (Initial Load)
CV.055 Implement Conversion Components (Initial Load)	CV.055 Implement Conversion Components (Initial Load)
Training	
TR.020 Prepare Project Team Learning Plan	TR.020 Prepare Project Team Learning Plan

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REQUIREMENTS ANALYSIS (RA) PROCESS GUIDANCE

RA.015 Develop Business Use Case Model

In this task, the business use cases that represent the business requirements are defined. Scenarios that further define the use case and the actors that interact with the system are also defined. In custom BI development, most of the use cases will be associated with obtaining and analyzing data. The development of a Business Use Case Model through the validation of the Functional Prototype (IM.010) relates to creating use cases that match business requirements to ensure the business intelligence data model can support all high priority business use case. This relates to the requirements modeling process in Data Warehouse Methodology.

RA.024 Develop Use Case Details

In Business Intelligence, there are many use cases to obtain data in order to perform analysis or create a report. Current process use cases may require obtaining data from multiple sources. This would actually provide at least two high level use cases for the BI environment; one to obtain the data from the multiple sources and integrate them into a BI environment, and the second use case documenting the business function that uses the data. Hence, the many "obtaining data" use cases are used to

create or extend the logical and physical data models. Business rules would be documented and incorporated into the Extract, Transform, and Load (ETL) processes documented in a tool such as Oracle Warehouse Builder (OWB). Some common use cases may wind up being stored procedures, derived fields, or materialized views in the final BI architecture. Of course, this is part of the analysis process that is common to all methodologies. Please note that the business processes will be used to ensure the data model can provide information to the business functions (obtained from the requirements) documented as use cases.

RA.030 Validate Conceptual Prototype

The validation usually occurs at a work session or the results of the work session may be validated via minutes from the work session with accompanying diagrams. It is not unusual for the Conceptual Prototype to be slightly altered as more detail requirements and system constraints are documented.

RA.035 Develop High-Level Software Architecture Description

In this task, you ensure the conceptual architecture can support the business requirements. As requirements in the form of use cases are better understood and prioritized, the architecture evolves from conceptual to a physical environment.

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ANALYSIS (AN) PROCESS GUIDANCE

AN.050 Analyze Data
AN.060 Analyze Behavior

In these tasks, you analyze use cases documented during requirements gathering and ultimate organize the results into the Analysis Specification. Use cases are associated with all the business and system processes (for example, adding a user to the environment, creating a purchase order, validating a payment was made to a particular vendor, etc.) These tasks are repeated for each use case, in the order of their priorities in the MoSCoW List (Must Have, Should Have, Could Have, Won't Have). Use cases can be classified into broad types such as interface classes, control classes, and data access classes. An example of a use cases in an interface class could be the inbound update use cases and the outbound extract use cases.

The Analysis Specification is the collation and organization of the Data Analysis (AN.050) and the Behavior Analysis (AN.060) contains the analyzed use case that show how the user requirements documented as use cases can be satisfied by the application. This can be done through diagrams, sequence of events, and user walk throughs. For example, data access requirements (use cases) would use the data model to show how data would be obtained for reporting or analysis. The requirement (use case) would then become a test case with expected results confirmed during user acceptance testing. This would complete the cycle from requirements through tested result using the use case concept.

AN.110 Review Analysis Model

Approach

These reviews can assist with learning events such as understanding the modeling approach when creating dimensional models. A new approach to analyzing data may generate tasks within the end-user training plan. Minutes from these review sessions are critical to tracking changes especially if the changes result in formal change requests.

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IMPLEMENTATION (IM) PROCESS GUIDANCE

IM.005 Develop Conceptual Prototype

The prototype diagrams are usually created during a work session and often documented in Visio. The diagram shows the interfaces to the data warehouse environment, how areas such as the Daily Business Intelligence might be used, and how the end user will access the environment.

The Conceptual Prototype will be expanded and is the first step in an iterative process to gain concurrence to the functional and technical architecture needed to support the requirements.

IM.040 Implement Database

The DDL to create the database is usually generated from a design tool such as Oracle® Warehouse Builder (OWB). In some cases such as Oracle® Siebel Analytics, the baseline physical model is extended to meet specific customer business requirements.

Implementing the database only generates the database structures and does not populate the tables with data. For BI, update processes must be created or configured in the case of prebuilt processes. Security at the data level such as Virtual Private Database (VPD) is usually included in the design.

IM.050 Implement Components

In many cases, the components can be generated using tools, such as Oracle Warehouse Builder (OWB). OWB can also be used as a design tool. The Data Mapping (CV.027) is a primary input to the ETL process.

IM.070 Assemble Components

An example of assembling components is creating a common updated script that calls multiple ETL modules to load a set of data. An example of data access may be using a Discoverer report as content for a portlet within a common portal area.

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TESTING (TE) PROCESS GUIDANCE

TE.005 Determine Testing Requirements

Review the Business and System Objectives (RD.001) and the Technical Architecture Workshop Results (TA.010) to understand the business and system objectives and the architecture of the solution.

The Testing Requirements define the business requirements for testing the data warehouse solution. It defines test requirements according to their significance or criticality to the business.

- Review the Technical Architecture Workshop Results (TA.010) to understand the technical risks in the solution. Some of them will become candidates for testing.
- Review Quality Management Plan (QM.010), and/or the Quality Control and Reporting Process (QM.020), as well as the Manage Issue and Problem Management Process for background information on Problem Management procedures.
- Work with ambassador users to identify the Testing Requirements and document them clearly. In the process, it may be required to understand the quality policy of the customer and try to conform to that.

Additional Prerequisites

The Technical Architecture Workshop Results (TA.010) contains requirements about the future technical architecture that should be reflected in the Testing Strategy.

TE.010 Develop Testing Strategy

Acceptance Criteria

One component of the Testing Strategy is specifying the Acceptance Criteria. In a custom BI engagement, the Acceptance Criteria defines the metrics required to obtain acceptance of the solution. It should include criteria to support successful implementation of the solution in production including:

- documentation
- extracted data
- data access
- data acquisition (ETL)
- system management
- solution administration

Test Scope and Approach (Objectives)

Unit Test

The objective of the Unit Test is to test the functionality of all the individual components that comprise the data warehouse solution, for example, ETL maps, error handling, report validations, calculations, security, data warehouse administration components, and adherence to standards. The Analysts typically perform this first level of testing in the data warehouse development environment during the Construction phase. As in all structured approaches to testing, the Testing Process emphasizes early testing (and verification in general) at each appropriate level. Adequate review of the design effectively prevents defects at the component level. Where the component is visible to an ambassador user the user should review and validate the component interactively. Where the output from the component is not visible, additional testing by the developer is needed before the component can be integrated.

System Test

The objective of the System Test is to verify the functionality, security and other system requirements of the entire data warehouse system. System testing may include regression testing to verify whether existing components of data warehouse solution delivered through earlier iterations work properly with the components of the new iteration.

Systems Integration Test

The objective of the Systems Integration Test is to verify the integration of the data warehouse system with the external applications. This will be carried out in the test environment. During this test, following interface components will be tested.

Final Data Validation and Data Access Test

This test is carried out in the Production Environment. This is a final walkthrough done by the development team before it is delivered to the users. All the business critical reports are tested in this step before going live.

Test Environments

There are several things to consider when defining the required test environments, such as:

- Review customer requirements for testing and release management for any specific environment requirements.
- Consider sources systems, for example, testing business scenarios in source to validate correct capture and processing in data warehouse may require the source system to be controlled.

Additional Prerequisites

The Architecture Requirements and Strategy (TA.020) contains requirements about the future technical architecture that may need to be reflected in the Testing Strategy.

TE.015 Develop Integration Test Plan

The purpose of this task is to produce a test plan for all points of integration of the data warehouse solution, excluding those to other (non-source) systems. The latter is covered in the systems integration test plan.

Test Roles and Responsibilities

Identify roles and assign responsibilities to development team members. List the roles and responsibilities involved in the use case testing of the data warehouse solution. Note that the roles listed in the OUM task may be too generic. You may have to change these roles based on the level of testing. Identify functional roles and assign to ambassador users. Identify ambassador users and group by functional area.

TE.020 Develop Unit Test Scripts

Checklists should be available for each type of the following components:

- Data Acquisition Components
- Data Access Components
- Administration
- Other Components

Depending on what component the developer is testing, he/she will have to use the corresponding checklist as guidelines for testing.

TE.050 Develop System Test Plan

Additional Test Types

Each system test type covers a class of technical, functional or non-functional requirements. Additional custom BI implementation test types include the following:

- System process sequencing using historical data
- Data reconciliation
- Database auditing
- Data archival

Features to be Tested

Categorize the test scenarios by the different aspects of the data warehouse (Data Acquisition, Data Access and Administration).

With regards to the test data:

- Ensure there is a process to validate that the data generated is correct.
- Ensure that the data is a representative sample.

Test Roles and Responsibilities

Identify roles and assign responsibilities to development team members. List the roles and responsibilities involved in the system testing of the data warehouse solution. Note that the roles listed in the OUM task may be too generic. You may have to change these roles based on the level of testing. Identify functional roles and assign to ambassador users. Identify ambassador users and group by functional area.

TE.080 Develop Systems Integration Test Plan

Task Steps (Step 3)

The Systems Integration Test Plan should cover all points of integration of the data warehouse solution that are related to other (non-source) systems.

In the Systems Integration Test Scope component, identify the external systems that the data warehouse feeds or provides other means of access. Identify key points of contact for each of these systems for further requirements and for resolving issues.

Test Roles and Responsibilities

Identify roles and assign responsibilities to development team members. List the roles and responsibilities involved in the system testing of the data warehouse solution. Note that the roles listed in the OUM task may be too generic. You may have to change these roles based on the level of testing. Identify functional roles and assign to ambassador users. Identify ambassador users and group by functional area.

TE.110 Prepare Acceptance Test Environment

As part of this task, you walk through business critical data and data access components to confirm that the system is ready to go into production. For example, data is validated to ensure that the numbers and calculations are correct; data access is confirmed to be appropriate for each user role. In some situations, when it is the first time the system is set in production, the Acceptance Test Environment may be the same as the Production Environment.

Task Steps

In a custom BI implementation, you have additional task steps 7 and 8 below:

No.	Task Step	Component	Component Description
1	Review the Testing Strategy (TE.010) to see what kind of testing environments should be used, and Acceptance Test Plan (TE.082) for more detailed requirements for the Acceptance Test Environment.	None	None
2	Determine the exact configurations required for the Acceptance Test Environment.	Acceptance Test Environment Setup Report	The Acceptance Test Environment Setup Report documents the exact configurations that will be/are used for the Acceptance Test Environment.
3	Determine required test data for the user acceptance test.	Test Data Baseline Manual Data Load Scripted Data Load Converted Data Load	The Test Data Baseline describes which test data baseline (from Prepared Test data (TE.018) will be used for the user acceptance test. The Manual Data Load describes any additional data that should be manually entered in the Acceptance Test Environment on top of the test data baseline. The Scripted Data Load describes any additional data that should be loaded into the Acceptance Test Environment on top of the test data baseline using scripts. The Converted Data Load describes any additional data that should be loaded into the Acceptance Test Environment on top of the test data baseline using conversion programs or scripts.
4	Determine required help data for the Acceptance Test	Help Text	The Help Text Data Load component describes all help text scripts or files must be

	Environment.	Data Load	included in the Acceptance Test Environment.
5	Determine which components should be included in the user acceptance test.	Application Object Load for Acceptance Test	The Application Object Load for Acceptance Test component describes all the sources that should be contained in the Acceptance Test Environment.
6	Set up physical Acceptance Test Environment.	Update of all components listed above	The components listed above will be updated along with the physical creation of the Acceptance Test Environment. The work product as a whole should be seen as a setup report that reflects how the Acceptance Test Environment actually has been set up.
7	Perform Business Critical Data Validation Test.	Business Critical Validation Test	This component lists the validation test results of the data components of the solution that hold business critical measures. The data components include stored measures as well as calculated measures.
8	Perform Data Access Capabilities Test.	Data Access Capabilities Test	This section records the results of the access capability testing that is performed to verify whether all the critical data access components (reports) are accessible by the appropriate users/roles as designed according to their access profiles.
9	Quality check Acceptance Test Environment.	Update of all components listed above	When the Acceptance Test Environment is ready to be used, it must be quality checked to see whether it works as expected. It might result in some changes/corrections, and the final work product must be updated to reflect any changes.

Approach

Verify the following:

- the correctness of the business critical data content shown by the reports by validating them
- whether the access capabilities of all reports are in place in the manner in which they are designed.

This is a critical task that is performed after the data is loaded in to the production data warehouse before the reports are made available to the users.

Business Critical Data Validation Test

When performing the Business Critical Data Validation you should identify the business critical data with the help of Information Analyst. This can be categorized by business or subject area. Also, identify the data access components (reports) that use these critical data elements. For each of these reports, do the following.

- Review the test scenario deliverables (TE.035 – Integration Test Scenarios, TE.025 - System Test Scenarios and TE.090 - Systems Integration Test Scenarios) to identify the scenarios intended to test this critical measure. Run the report with the parameters taken from the script and note down the value shown by the report.
- Validate the correctness of the data/calculation by verifying the report result against the expected value. The expected value is documented in the test scenarios.
- In cases where identified test scenarios cannot be used for the validation test, create test cases of your own and document the steps in the column ‘Parameters for Validation’ in the template.
- In cases where test scenarios can be used, make a reference to the test case ID in the column ‘‘Parameters for Validation’.
- Record the results and update the status as pass/fail

When you have done this, you should consider the following:

- Have all data elements been verified?
- Is the list of elements complete?
- Has customer overlooked any key elements which should be included?

Data Access Capabilities Test

When performing the Data Access Capabilities Test you should first understand the user profiles and role access definitions. Then list the critical data access components (reports) and list the user roles that have been provided access to this report. For each of these reports:

- Test whether the report is accessible for this role. This can be performed by creating a dummy user and then assigning the user this particular role and then logging in as the dummy user and verifying if the report is accessible.
- If the Fine grained Security solution is used in the project (for example, VPD) fill up the column 'Data Valued allowed by the role' with the textual description or range of data. Delete this column, if this is not applicable to your project.
- Record the results.

When you have done this, you should consider the following:

- Are elements accessible to reporting users?
- Are tables and views accessible to the reports?

Additional Prerequisites

The Converted and Verified Data (CV.065) is the data that is loaded into the Acceptance Test Environment to identify and define test types, test items and features to be tested.

Audience, Distribution and Usage

This work product is intended to serve as a record for the data validation test and access capabilities test.

Additional Quality Criteria

- Have all business critical data elements been identified and validated for correctness?
- Have the access capabilities of reports or other components been defined per the User Role access document?

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PERFORMANCE MANAGEMENT (PT) PROCESS GUIDANCE

PT.030 Define Performance Testing Strategy

Depending on your engagement, sometimes the Performance Testing Strategy can be included as section in the overall Testing Strategy (TE.010) work product.

PT.120 Conduct Production Performance Management

Typically, in a custom BI implementation, this is part of standard performance monitoring and is accounted for in administrative modules. Usually we recommend monitoring tools and do not get directly involved with ongoing monitoring unless there are specific requirements for monitoring performance.

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DATA ACQUISITION AND CONVERSION (CV) PROCESS GUIDANCE

CV.010 Define Data Acquisition and Conversion Requirements

Approach

Data Acquisition - Source System Constraints - System Interdependencies Component

When the analysis is complete, create a System Interdependencies diagram. It should identify all systems considered as data sources for the data warehouse with a clear indication of the main data flows and dependencies.

The diagram is used to decide which of the systems should be used as a source for a particular entity. It is also used to check completeness of sources and to resolve conflicts.

CV.020 Define Data Acquisition, Conversion and Data Quality Strategy

Task Steps

The following task step is specific to custom BI:

No.	Task Step	Component	Component Description
BI	Provide guidelines for refreshing different types of objects. Include only those objects that are appropriate for the solution. Optionally, prepare templates for developers for the common tasks (for example, template for refreshing SCD type 1 dimension, another template for refreshing SCD type 2 dimension, another template for loading monthly partitioned fact table).	Data Acquisition - Data Load Strategy - Data Refresh Strategy	The Data Refresh Strategy component describes strategies for updating the data, including: • Referential Data Refresh Approach • Dimensional Table Refresh Approach • Fact Table Refresh Approach • Summary Tables Refresh Approach • OLAP Dimensions Refresh Approach • OLAP Cubes Refresh Approach Delete the sections that are not appropriate.

Task step 8 has additional information specific to custom BI:.

No.	Task Step	Component	Component Description
8	Review the business requirements regarding the initial and historical data in order to document each requirement. Get the sign off from the customer for the initial and historical load requirements. Custom BI - Make sure that both the historical loads and initial loads are included (i.e. list everything that has to be loaded before the data warehouse can be refreshed on regular basis with the current data).	Data Acquisition - Data Load Strategy - Initial and Historical Loads	The Initial and Historical Loads component provides a table listing the requirements for the initial or historical loads from source systems. It is important to note for each requirement how many months (days) of data have to be loaded, whether the historical data is available, what is the expected volume of data and whether it is possible to load the historical data with the processes developed for regular updates.

Approach

Data Acquisition - Data Load Strategy - Data Refresh Strategy

Provide guidelines for refreshing different types of objects, including:

- Referential Data Refresh Approach – how and when to update referential data (e.g. list of values) that are not provided as part of source system interfaces, etc.
- Dimensional Table Refresh Approach – how to update dimensions, detect changes in dimensions, take care of slowly changing dimensions, etc.
- Fact Table Refresh Approach – how to update fact tables, maintain partitions and indexes, publish newly loaded partitions to users, etc.
- Summary Tables Refresh Approach – how and when to refresh summary tables, how to maintain partitions of summary tables, etc.
- OLAP Dimensions Refresh Approach – how and when to update OLAP dimensions and hierarchies from the data warehouse data, etc.
- OLAP Cubes Refresh Approach – how and when to update OLAP cubes from the data warehouse facts, how to rollup cubes, etc.

Data Acquisition - Data Load Strategy - Referential Data Refresh Approach

The best practice it to maintain the referential data (for example, list of values) from source systems as any other data. However, this is not always possible since the referential data might be hardcoded in source systems, the interface for referential data might not exists or there is simply no source system for a particular referential data. Therefore, document how the referential data is handled in the solution, how they are updated and who is responsible for the update. Consider creating a special area to store the referential data and use the area as if it was any other source system.

Data Acquisition - Data Load Strategy - Dimensional Table Refresh Approach

Provide the guidelines on refreshing dimensions. Include methods for detecting changes in dimensions, methods for implementing slowly changing dimensions and methods to update audit attributes.

A typical method for detecting changes is to compare the source data with the data in the data warehouse using set operations (minus, union) in order to prepare data set containing only the changes. This method can be used with both full and incremental extracts.

Data Acquisition - Data Load Strategy Component - Fact Table Refresh Approach Section

Provide the guidelines on refreshing fact tables. Include methods for maintaining table partitions and indexes, how to publish the newly loaded data to users and methods to update audit attributes.

Consider using the Partition Exchange Loading (PEL) method when updating fact tables. PEL is the preferred method for updating partitioned fact tables. Decide whether to use out of the box PEL support in OWB (with all its restrictions) or a custom approach.

If the system allows backward changes to fact tables describe methods how to handle and propagate the changes (for example, if it is necessary to load late arriving transactions, how to insert the transactions to a fact table, how to differentiate late arriving records from the other records, how to propagate late arriving records to summaries).

Data Acquisition - Data Load Strategy Component - Summary Tables Refresh Approach Section

Provide the guidelines on refreshing summary tables. Include methods on how to design the summary tables, maintain partitions and refresh the summaries.

Data Acquisition - Data Load Strategy Component - OLAP Dimensions Refresh Approach Section

If OLAP area is part of the solution, provide guidelines how to update OLAP dimensions and hierarchies from the data warehouse.

Data Acquisition - Data Load Strategy Component - OLAP Cubes Refresh Approach Section

If OLAP area is part of the solution, provide guidelines how to update OLAP cubes from the data warehouse. Include methods on updating the cubes and performing rollups.

The following boilerplate text should be reviewed and copied, if appropriate, into the corresponding section, Exception Handling, of the work product template for this task.

Note that not all boilerplate text is applicable, and additional materials might be needed for the final customer work product.

Data Acquisition - Data Load Strategy Component - Error Hanndling Strategy - Error Notification Approach Section

If using OWB, consider using notifications raised from Oracle Workflow. These notifications can be configured in almost any way to suite data warehouse needs. Administrators can reply to Oracle Workflow notifications to stop or resume the process, users can provide confirmations through the notifications etc.

Data Quality - Objectives Component - Data Quality Overview Section

The following boilerplate text might be used to describe the data quality issues in the data warehouse. This text can be added to the Data Quality Overview section of the work product.

The best way how to answer “why is the data quality important” is to reverse the question - what happens when the data do not have required quality. There are many cases of incurred costs, lost credibility or missed business opportunities because of incorrect data. For example

- Sending direct mails to incorrect addresses
- Contacting dead customers
- Contacting the same customer twice with the same offer
- Offering products to customers that should not qualify
- Refusing products to customers that should qualify
- Restatement of financial statements
- Wrong data sent to regulators

A data warehouse is especially vulnerable to poor data quality. Unlike legacy systems, the primary objective of the data warehouse is to provide data to knowledge workers and to analytical applications. If the data quality does not allow knowledge workers to use it for analysis or decision making they will stop using the system and it will fail.

A data warehouse is also usually the first system that consolidates data from many sources into a common data model. As a result it is much easier to discover the data quality issues in the data warehouse than it is in legacy systems.

So even if the data warehouse does not cause most of the data errors, it is in the vital interest of the data warehouse project to influence the source systems and information producers to improve the data quality.

There are many reasons why data can be incorrect. The following are some typical reasons for a data warehouse environment, ordered by the information flow sequence.

- Information is incorrectly provided to data entry clerks.
- Information is incorrectly entered into a source system.
- Optional data are not entered.
- Source system requires information that is not available.
- Information is duplicated in the source system.
- The same information in different source systems is inconsistent.
- The source system incorrectly calculates derived data.
- The data are incorrectly extracted from the source system.
- The data are extracted from source systems at bad time.
- The files are incorrectly transferred to the data warehouse.
- The data are incorrectly processed in the data warehouse.
- Users select incorrect data from the data warehouse.
- Users do not understand data in the data warehouse.
- Behavior of source systems changes over time but the rules and processes are not updated to reflect this change.

The previous paragraph clearly shows that the data quality is not a function of one system or one role. Instead, all the applications, architects, information producers and knowledge workers have to take responsibility for continually improving the data quality.

Therefore it is necessary for an organization to establish an environment that leads to sustainable improving of information quality. The data quality initiatives might bring some short term benefits but unless it is a continuing process the data quality will deteriorate over time.

Data Quality - Processes Component - Data Transfer Section

The following methods might be considered for ensuring the data transfer integrity for the specific data warehouse solution that should be included in the work product template for this task.

One method how to control the data transfer process is via “control files”.

Control files complement the data files. They contain information about data files within a file set, version of the file structure, effective date, number of records in the data files and (if no character conversion takes place) they could also contain number of characters and file checksum.

When data files are received and before any processing is done, the files are checked against the control files. If the data file does not correspond with the control file than the administrator is notified, the whole data file is considered corrupt and either the transfer or the data extraction must be repeated.

Another method is to structure the data files in a way that enables consistency checks of the files. The following rules can be used:

- Each row is prefixed with a row type
- One row type correspond to the file header, containing
    - File name
    - Effective date
    - File structure version
- One row type correspond to the file footer, containing
    - Number of data rows within the file
    - Optionally other summary measures (e.g. sum of balances)

When data files are received and before any processing is done, the file structure is checked. The file must have the header row and the footer row, the effective date and file structure version should correspond to the expected values, the number of data rows and summary measures have to be correct. If the file does not meet the checks the whole file or file set is rejected.

All these methods can be used together to provide a maximum level of control over the data transfer process.

Data Quality - Processes Component - Audit and Control Section

The following text and example may be useful as a starting point for defining the audit and control methods for the specific data warehouse solution that should be included in the deliverable template for this task.

The following text describes where to put automatic the data quality checks:

Checkpoints in the ETL Process are automatic checks executed during the extraction, transformation and load process (ETL) that validate number of processed records. If the number of records does not correspond the administrator is notified and optionally the whole process is stopped until the problem is resolved. The typical checkpoints are

- After Extraction – check number of extracted records.
- After Transfer – check number of records before and after the transfer.
- After Load – check number of records loaded.
- After Allocation – check sum of allocated figures against the total.
- After Aggregation – check the sum of detail data against the aggregation.

Number of records can be checked against the source (in case of 1:1 mappings), or, alternatively, it is possible for each data file to define a typical volume of data and the check is done against this volume. For example, we can define that the incremental extract should have between 0 and 50000 records or that the full extract should have

more than 4 million records. If these conditions are not met than the administrator is notified.

Usually if the data does not meet these quality checks than the whole data set is considered corrupt and the ETL process is stopped until a detailed analysis is done and the process is either resumed or canceled.

The following methods can be used to report the data quality and to inform about the data (non) quality in the data warehouse:

Method	Description
Exception Reports	Exception reports list the detail records that contain missing values or do not conform to the business rules. The exception reports are usually passed to process or data owners for correction. Exception reports are usually produced ex-post, i.e. after the data is loaded into the data warehouse. It is therefore necessary for the system to re-process the corrected values after they are repaired in the source system. If a number of records to be corrected is not large, than the exception records should be produced and distributed automatically. However, if the number of records to be corrected is significant, it is better to run a focused activity targeted at the quality improvement and only after the number drops to acceptable level to produce automatic exception reports. For example, exception records can list active accounts with obsolete product, accounts with invalid interest rates or customers who should be merged with another customer (potential duplicates).
Summary Reports	Summary reports show percentage of records that meet quality criteria. Summary reports are dimensioned by source systems, products or organizational unit, in order to better understand the distribution of errors. Summary reports provide the basis for regular monitoring of data quality. Similar to exception reports, summary reports can be defined only if the quality measures can be calculated using the data in the data warehouse. Summary reports are produced automatically. It is useful to store results of summary reports in the database to support evaluation of data quality over time. For example, summary reports can show percentage of correct addresses per address type and source system, percentage of duplicate customers per customer type or percentage of non classified current accounts in debit. Summary reports can also be used to measure a completeness of data – a typical report will show percentage of records that do not have valid values per table column and source system. This report is also quite useful for knowledge workers.
Reconciliation Reports	Reconciliation reports compare data from the data warehouse with the same data from another surrogate data source. The reconciliation reports are used to measure difference between these two systems. A typical example is to measure summary balances on customer accounts against the balances on GL accounts. Also it is possible to compare reports from source systems against the reports produced from the data warehouse.

CV.027 Perform Data Mapping

This task evolves during the development of a project. It begins with the identification of the source systems and evolves into the actual data source of the attributes needed in the target system. In many cases some type of transformation is needed to conform to the target system's architecture. A byproduct of this task is the gap analysis matrix. Some data elements needed in the target system may not be available. These element may be entered manually through the application or in some cases a new function is required to maintain the "missing" source data elements. The final source to target mappings with the transformations may be documented in a tool such as Oracle Warehouse Builder (OWB).

CV.040 Design Conversion Components (Initial Load)

Approach

For BI Data Acquisition, the primary objective of this task is to describe the specific ETL processes that enable the interfaces between source and target systems. It identifies all components of each ETL process including data complexities or constraints for conversions and interfaces. In addition test scenarios or cases are developed to support unit testing of each data acquisition component.

When the project team performs this task, most of the targets would have been designed and their mappings to source system objects identified (See CV.027, Data Mapping). High-level requirements and constraints of the interfaces would also be available from the the Data Acquisition, Conversion and Data Quality Strategy (CV.020). All components of the solution would be identified in the Final Platform and Network Architecture (TA.150).

This task involves understanding these requirements and translating them to specifications for individual mappings that extract, transform and load data between systems, and how they would fit in the overall architecture.

As far as possible, the deliverable template has been designed to make it independent of the ETL implementation tool, be it using custom PL/SQL scripts or using Oracle Warehouse Builder.

<Mapping> Specification Component - Exception Handling Section

Error Handling Types

- The handling types are put into context in the actual mapping descriptions.
- <Adjust the suggested list to the actual project context>
- Code Description
- EH1 Load partial data into DWH, log received sources in log table
- EH2 Don't load data into DWH, log data in error table with "wait" status
- EH3 Load data into DWH and log problem into log table
- EH4 Load data into DWH and add the missing dimension to the dimension table with description ‘UNKNOWN’. Log missing dimension into log table.
- EH5 Handle the received data with a clear status to the users
- <…>

Error Correction Types

- The error correction types are put into context in the actual mapping descriptions.
- <Adjust the suggested list to the actual project context>
- Code Description
- EC1 Load missing transaction(s) during next normal program run
- EC2 Load missing transaction(s) by explicit program run
- EC3 Update the missing dimension table with the correct dimension information from source during next normal run
- EC4 Allow manual update of data in the DWH directly (by user through predefined application; by IT people)
- EC5 Correct data automatically according to pre-defined business rules
- EC6 Update DWH data with new version during next normal program run.

CV.055 Implement Conversion Components (Initial Load)

Task Step

The following task step is specific to custom BI:

No.	Task Step	Component	Component Description
3	Provide the ETL developers with a checklist against which they can record the progress of ETL component development.	Build and Unit Test ETL Components	You use the information captured in the Conversion Component Designs (CV.040) to build the ETL components. This component provides a checklist for the ETL developers and testers.

Approach

ETL Components

By this time in the project the information on data source systems, data transformation rules and target system data model design is available. The project team has outlined the steps for the extraction, transformation, and loading of data for the developers, and the toolset(s) to be used for the process development.

Information on all the data extraction interface components, data transportation mechanisms, change data capture methods and constraints with respect to these have already been captured in the Conversion Component Designs (CV.040).

ETL Build Checklist

Provide the ETL developers with a checklist against which they can record the progress of ETL component development. Use the Conversion Component Designs (CV.040) to create a checklist table (basically, a list of ETL components that must be built) with the following columns:

- No. - Use CV.040 to obtain this information.
- Data Flow - Use CV.040 to obtain this information.
- Mapping Name - Use CV.040 to obtain this information.
- ETL Technology
- Built By - ETL developer/programmer name
- Checked By - Supervisor name
- Date - Date code ready for unit testing

Keep in mind the following:

- The precise detail for constructing the ETL Component is specified in the Conversion Component Designs (CV.040) together with the suggested ETL technology.
- Code should be deemed ready for unit testing when the developer is satisfied that they have covered all of the specified requirements.
- These paper based tracking mechanisms should be used in addition to any tool repository based means.

ETL Unit Test Checklist

Provide the ETL testers with a checklist against which they can record the progress of ETL component unit testing. (See also TE.030, Perform Unit Test). Create a second tabular checklist table similar to the ETL Build Checklist (basically, a list of ETL components that must be built) with the following columns:

- No. - Complete the same as the ETL Build Checklist.
- Data Flow - Complete the same as the ETL Build Checklist.
- Mapping Name - Complete the same as the ETL Build Checklist.
- Test Case Name - Case Name
- Tested By - Tester name
- Unit Test Result - Pass / Fail
- Date - Date unit tested
- Comment

Keep in mind the following:

- A component may have more than one test case associated with it.
- The purpose of designing and running a test is to find errors. Only by finding and correcting defects can the team build user confidence in the system.
- Someone other than its creator should test a component. Independent testing is more effective than testing performed by the author or developer.
- Tests must be repeatable. Therefore, it is necessary to document what has been tested. There might also be a requirement to retain a record of each test for audit purposes.
- The resulting work product serves as evidence of testing.

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TRAINING (TR) PROCESS GUIDANCE

TR.020 Prepare Project Team Learning Plan

Approach

Many times, training of customer staff can be incorporated into normal activities. For example, customers new to business intelligence may be familiar with normalized data models but may not be familiar with dimensional modeling. Design reviews can incorporate learning activities. Also, customer developers may assist by developing modules using tools such as Oracle Warehouse Builder. Ensure the customer developers get training on new tools and account for their newly acquired skills in the project plan. Ideally, the customer developers should reduce the time needed for development but be sure to factor in the need for Oracle consultants to help customer developers.

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